# SEALS User Guide

## Spatial Economic Allocation Landscape Simulator

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# 1 Introduction

SEALS, the Spatial Economic Allocation Landscape Simulator, is a land-use change model that downscales predictions of land-use change from aggregate (regional or coarse-gridded) inputs to a finer resolution, typically 10–300 meters.

SEALS' primary comparative advantage is fast computation. It downscales to 300 meters globally — roughly 8.4 billion grid cells — in about an hour on a laptop. It is designed with parallelization in mind and so scales well to high-performance, cloud, or distributed computing. Performance-critical functions are written in C++ (via Cython); everything else is Python.

## 1.1 How to read this guide

Start at Installation and Quickstart — together they take you from nothing to a downscaled map. Then:

- **Walkthrough** — a guided first run, and the anatomy of a run file. Read this before you start a project of your own.
- **Custom scenarios** — the column reference for the scenarios CSV that defines what SEALS runs.
- **MAgPIE** — coupling SEALS to an external land-use model, which is also the general pattern for coupling it to any coarse projection source.

This guide is rendered both as a website and as a single PDF (download). If you are reading the PDF, links to the other two products point at the published site.

## 1.2 Integration with the Earth-Economy Devstack

SEALS is one of three software products maintained by NatCap TEEMs (The Earth-Economy Modellers):

| product | what it is |
|---|---|
| **SEALS** | this model — the downscaler |
| **GTAP-InVEST** | the coupled economy–ecosystem model that uses SEALS for its land-use step |

| product | what it is |
| --- | --- |
| **Earth-Economy Devstack** | the shared platform: hazelbean, ProjectFlow, and the conventions all three follow |

To learn about the rest of the Devstack, see its documentation at https://justinandrewjohnson.com/earth_economy_devstack/.

# 2 Installation

SEALS is installed directly from the Git repository as an *editable install* in a conda environment.

## 2.1 1. Install git and Miniforge/Conda

If not already installed:

- **git** — https://git-scm.com/downloads. The install starts from a clone.
- **Miniforge** — https://github.com/conda-forge/miniforge. Install for your user account only, and accept the option to add it to your `PATH`.

## 2.2 2. Install a C/C++ compiler

SEALS compiles a Cython extension (`seals/seals_cython_functions.pyx`) when it is installed from source, so **a C/C++ compiler is required** — see Compiling C/C++ Code.

- **macOS:** `xcode-select --install`
- **Windows:** Visual Studio Build Tools, or run `install.bat` from the Earth-Economy Devstack repository root
- **Linux:** `gcc` is usually already present

## 2.3 3. Create and activate an environment

```
conda create -n <your_env>
conda activate <your_env>
```

Pick your own environment name — the devstack does not require a particular one. Always activate it before running SEALS scripts.

The environment starts empty; Python arrives in step 4 as a dependency of hazelbean, at a version conda knows hazelbean was built for.

## 2.4 4. Install hazelbean and the dependency stack from conda-forge

This is the step that makes the editable install below work, so do not skip it — it is where hazelbean, GDAL, NumPy and the rest of the geospatial stack come from, as conda builds rather than pip wheels:

```
conda install hazelbean cython libgdal-hdf5
```

This may take 5–10 minutes.

## 2.5 5. Clone SEALS into the devstack layout

You may clone SEALS wherever you want, however documentation here assumes the repo lives in its own project folder under `~/Files/` (`C:\Users\<you>\Files\` on Windows). The command below clones it into a `seals` subfolder in files.

```
mkdir -p ~/Files/seals && cd ~/Files/seals
git clone https://github.com/jandrewjohnson/seals.git
```

## 2.6 6. Make the editable install

Note that the second `seals` is the actual repository. Projects will and all written files will go in the first `seals`.

```
cd ~/Files/seals/seals
pip install -e . --no-deps
```

`--no-deps` is deliberate, and required. SEALS declares `hazelbean>=2.0.0` as its only runtime dependency, and you already have it from step 4. Letting pip resolve dependencies again tends to replace conda's builds of the geospatial stack (GDAL and friends) with incompatible wheels.

## 2.7 8. Other devstack repos (optional)

The same clone-and-editable-install pattern works for the rest of the stack, each in its own `~/Files/<name>/` project folder: global_invest_dev, gtap_invest_dev, gtappy_dev, gep_dev. All but `global_invest_dev` are private; ask for read access before cloning them.

See the full installation guide for the devstack-wide version of these steps.

## 2.8 Troubleshooting

The devstack guide's Common problems section covers the environment-level failures, none of which are SEALS-specific: you need administrator rights; PowerShell needs `conda init powershell` before conda works in it; conda may need adding to `PATH` manually; and Windows may show “Windows protected your PC” on the Miniforge installer (click *More info* → *Run anyway*).

If `import seals` resolves to `site-packages`, see the verify step in 7 — an installed copy is shadowing your clone.

## 2.9 Next

With the environment activated, go to Quickstart — it runs the pared test pipeline and shows where the output lands.

Budget time for the first run rather than the install: SEALS downloads the base data it needs at runtime from a public bucket, so a correct install is still followed by a substantial download before you see a result. No credentials are needed for the default configuration.

# 3 Quickstart

Get SEALS running, then make it your own. This picks up where Installation leaves off — it assumes you have an activated environment with `hazelbean` from conda-forge and SEALS editable-installed from your clone. For what the model actually does, see Introduction.

## 3.1 1. Run it

Two run files ship with SEALS, in `seals/seals/`:

| file | what it runs |
|---|---|
| `run_seals.py` | the standard pipeline: three scenarios, 2030 and 2050 |
| `run_seals_test.py` | the same pipeline pared down — baseline plus one BAU, one year, Rwanda |

Start with the test:

```
conda activate <your_env>
cd seals/seals
python run_seals_test.py
```

Required base data is downloaded at runtime from a public bucket; no credentials are needed for the default configuration.

## 3.2 2. Where the output goes

Not into the repo. Because the run file lives inside a cloned repo, ProjectFlow places the project directory just *outside* it:

```
~/Files/seals/projects/seals_test/
  input/          seeded from the repo's input_template/
  intermediate/   one folder per task
  output/
```

Run the same command a second time and it finishes almost immediately — every task is behind an existence check, so completed work is skipped. That is the main thing ProjectFlow buys you: interrupt a long run, fix one input, re-run, and only what is missing recomputes.

## 3.3 3. Make it your own

Copy the run file into your own project repo as `run_<project>.py` and change the two lines in its `__main__` block — that is the whole of the configuration:

```
if __name__ == '__main__':
    p = hb.ProjectFlow(project_name='my_project', run_mode='check')
    p.scenario_definitions_filename = 'my_project_scenarios.csv'

    run_project(p)
```

`run_mode` controls how much prior work is reused: `'check'` (the default) resumes in a stable project dir; `'full'` timestamps a fresh one.

Put your scenarios CSV in an `input_template/` directory beside the run file. It is tracked in git, and ProjectFlow copies anything missing into the project's untracked `input/` on first run without ever overwriting your working copy. The columns are documented in Custom scenarios.

To run the same pipeline a second way — a smoke test, a different AOI — write a *second run file* that imports the first rather than copying it. `run_seals_test.py` is the worked example, and the pattern is explained in conventions.

## 3.4 Next

- Walkthrough — a guided tour with screenshots
- Conventions — the full run-file rules
- Project complexity — why run files are shaped this way

# 4 Walkthrough

## 4.1 Getting set up

- Follow the installation page first.
- This guide assumes you installed SEALS using the devstack layout — its own project folder under `~/Files/`, with the repo inside it at `~/Files/seals/seals`.

## 4.2 Explore the SEALS code

- The repo root `seals/` holds more than the library — docs, tests, scripts
- The library itself is the `seals/seals` subdirectory. The nesting looks redundant but is how Python packaging works
- Inside it, `__init__.py` is what makes the directory importable as a package
- The modules are split by what they hold:
  - `seals_main.py`, `seals_tasks.py`, `seals_generate_base_data.py` — task functions, the actual model logic
  - `seals_initialize_project.py` — the task-tree builders
  - `seals_utils.py` — helpers
  - `run_seals.py` — the run file

## 4.3 Run files

- You do not run `seals_main.py` directly
- A **run file** configures a project and executes a task tree built from the library's task functions
- Two ship with SEALS, both in `seals/seals`:
  - `run_seals.py` — three scenarios, 2030 and 2050
  - `run_seals_test.py` — the same pipeline pared to a baseline, one BAU scenario, one year, Rwanda. Start here
- Open `run_seals.py`. It is about 60 lines and has three parts

## 4.4 Part 1: the task tree

- `build_task_tree(p)` answers one question: *what is this pipeline?*
- It contains nothing but tree construction. For stock SEALS it is one line, delegating to the shared library builder
- Compose extra library subtrees or your own tasks here if your project diverges

```
def build_task_tree(p):
    seals_initialize_project.build_standard_task_tree(p)
```

## 4.5 Part 2: run_project

- `run_project(p)` answers *how is it executed?* It takes only `p`
- The rule that decides what goes where:
  - **`run_project(p)` sets what no variant ever changes** — base data location, processing resolution
  - **the caller sets what a variant might** — project name, run mode, scenarios CSV
- When a constant starts varying, it moves one line up into the caller. There is no signature to edit and no default to keep in sync

```
def run_project(p):
    p.run_in_parallel = 1         # before the tree: it has parallel iterators
    build_task_tree(p)
    p.skip_tasks(p.tasks_to_skip)

    p.base_data_dir = os.path.join(p.user_dir, 'Files', 'base_data')
    p.processing_resolution = 1.0

    hb.initialize_scenarios(p, p.scenario_definitions_filename)
    seals_initialize_project.initialize_project(p)   # seals' model initializer:
    # advanced options + derived attributes; must come AFTER the scenarios load

    p.execute()
    return p
```

## 4.6 Part 3: the __main__ guard

- This is where the project is configured — two lines of actual choice
- The guard is mandatory: importing a run file must never start a run. That is what lets a second run file import this pipeline instead of copying it

```python
if __name__ == '__main__':
    p = hb.ProjectFlow(project_name='seals', run_mode='check')
    p.scenario_definitions_filename = 'standard_scenarios.csv'

    run_project(p)
```

## 4.7 The p object

- `hb.ProjectFlow()` is a class — a recipe for an object. Calling it produces an object, which we assign to `p`
- `p` carries the project's **attributes**: `p.base_data_dir`, `p.scenario_definitions_filename`, and so on
- It also has **methods** that act on it: `p.add_task()`, `p.skip_tasks()`, `p.get_path()`, `p.execute()`
- Every task receives `p` and reads its configuration from it. That is how configuration set in `__main__` reaches code deep in the library

## 4.8 Where the project directory goes

- Not into the repo. Because the run file sits inside a cloned repo, ProjectFlow places the project dir just *outside* it, at `~/Files/seals/projects/<project_name>/`
- You do not compute this path yourself, and you should not save data in `seals/seals`
- `run_mode` decides how much prior work is reused:
  - `'check'` — reuse the stable dir, recompute only what is missing (the default)
  - `'fresh_intermediate'` — redo all computation, keep `input/` (test projects only)
  - `'full'` — a fresh timestamped dir; everything re-runs

# 5 Scenario definitions

## 5.1 The scenarios CSV

- The scenarios CSV specifies the runs you want. **Each row is one scenario**
- As the model iterates, it re-hydrates `p` from the row it is on — which is why scenario-varying values belong in the CSV and not in the run file
- The tracked copy lives in `input_template/` beside the run file. ProjectFlow copies anything missing into the project's `input/` on first run and never overwrites your working copy
- If the named file is nowhere to be found, SEALS generates a default so a first run still works

```
p.scenario_definitions_filename = 'standard_scenarios.csv'
```

## 5.2 Scenario types

- `scenario_type` is a **column in the CSV**, not something a run file sets
- `baseline` — the observed anchor; not downscaled, supplies base-year LULC
- `policy` — a trajectory to downscale, compared against a counterfactual
- `bau` — business-as-usual, being deprecated in favour of `baseline`; it still appears in shipped CSVs
- Full column reference: Custom scenarios

## 5.3 Automatically downloading data

- Paths in the CSV are **ref_paths** — relative locations resolved by `p.get_path()`, which searches the project `input/` dir, then `base_data`, then a cloud bucket, downloading what is missing
- NOTE: The first run downloads a lot of data. The default bucket is public and requires no credentials, but it is large and the download may take hours. Also note that you may need to restart the run file after it all downloads.

- `p.base_data_dir` is where those downloads land. Point it somewhere with room for very large files; the same base data serves every project
- The directory must be named `base_data` to match the bucket convention
- The default bucket is public — no credentials needed

```
p.base_data_dir = os.path.join(p.user_dir, 'Files', 'base_data')
```

# 6 Running it

## 6.1 Run the model

```
conda activate <your_env>
cd seals/seals
python run_seals_test.py
```

- On startup SEALS prints the **task tree** it is about to compute
- To understand the model in depth, read the task functions behind those names
- Run the same command a second time: it finishes almost immediately, because every task is behind an existence check and completed work is skipped

## 6.2 Find the output

- Go to `~/Files/seals/projects/seals_test/intermediate/`
- There is one directory per task in the tree
- The downscaled maps are in `stitched_lulc_simplified_scenarios/`, for example:

```
lulc_esa_seals7_ssp2_rcp45_luh2-message_bau_2030.tif
```

- Open it in QGIS alongside the base-year LULC and compare — that is your high-resolution land-use change projection

## 6.3 Next

SEALS is customized be adding or changing scenarios in the scenarios CSV. See Custom scenarios for the rules and conventions.

# 7 Custom Scenarios

The scenarios CSV defines your SEALS scenarios. Each row is one scenario. The tracked copies that ship with SEALS are `seals/input_template/standard_scenarios.csv` (three scenarios) and `standard_scenarios_test.csv` (a pared two-row version); this page documents their columns.

> **This is the v1 schema — the one the code reads today.** A **v2 schema** is specified in proposed_changes.qmd and no code implements it yet. v2 fixes several structural problems in v1: it replaces `baseline_reference_label` / `comparison_counterfactual_labels` / per-model time columns with a `observed_reference_label` / `compared_to` / `depends_on` split and a single canonical clock, and it allows hindcasting. Write v1 for now; expect a migration when the hazelbean parser lands.

## 7.1 Column reference

Values in the Example column are taken from the shipped `standard_scenarios.csv`.

| Column | Description | Example |
|---|---|---|
| `scenario_label` | Unique scenario identifier | `baseline_luh2-message`, `ssp2_rcp45_luh2-message_bau` |
| `scenario_type` | `baseline`, `bau`, or `policy` — see the note below | `baseline` |
| `aoi` | Area of interest: `global`, an ISO3 code, or a path to a vector | `RWA` |
| `exogenous_label` | Exogenous driver set | `baseline`, `ssp2` |
| `climate_label` | Climate scenario | `rcp45` |
| `model_label` | Coarse-projection model source | `luh2-message` |
| `counterfactual_label` | Counterfactual this row represents (blank on the baseline row) | `bau`, `bau_shift` |
| `years` | Space-separated projection years | `2030 2050` |

| Column | Description | Example |
|---|---|---|
| baseline_reference_label | scenario_label of the baseline row this one is measured against (blank on the baseline row) | baseline_luh2-message |
| base_years | Observed/base year(s) | 2017 |
| key_base_year | The base year used for LULC | 2017 |
| comparison_counterfactual_labels | Scenario(s) this row is compared against in reporting | ssp2_rcp45_luh2-message_bau |
| time_dim_adjustment | Offset applied to the source time dimension | add2015 |
| coarse_projections_input_path | Coarse projections NetCDF | luh2/raw_data/rcp45_ssp2/multiple-states_..._2015-2100.nc |
| lulc_src_label | Fine LULC source | esa |
| lulc_simplification_label | Fine LULC class scheme | seals7 |
| lulc_correspondence_path | Fine class correspondence | seals/default_inputs/esa_seals7_corre |
| coarse_src_label | Coarse data source | luh2-14 |
| coarse_simplification_label | Coarse class scheme | seals7 |
| coarse_correspondence_path | Coarse class correspondence | seals/default_inputs/luh2-14_seals7_correspondence.csv |
| lc_class_varname | Land-class variable in the NetCDF | all_variables |
| dimensions | Dimensions to read | time |
| calibration_parameters_source | Downscaling coefficients | seals/default_inputs/default_global_c |
| base_year_lulc_path | Base-year LULC raster | lulc/esa/lulc_esa_2017.tif |
| regional_projections_input_path | Regional projections CSV (optional; blank if coarse-only) | seals/default_inputs/rwa_bdi_regional |
| regions_vector_path | Regions vector | cartographic/ee/ee_r264_correspondenc |
| regions_column_label | Region column in that vector | ee_r264_label |

Paths are **ref_paths**: relative locations resolved by `p.get_path()`, which searches the project `input/` dir, then `base_data`, then the cloud bucket, downloading if needed. Always write them with **forward slashes**, on every platform.

### 7.1.1 A note on `scenario_type`

Three values appear in current files:

- `baseline` — the observed anchor. Not downscaled; it supplies base-year LULC and calibration inputs.

- `bau` — a business-as-usual trajectory. **Deprecated in favour of `baseline`.** The migration is in progress and unevenly applied, so `bau` still appears in shipped CSVs and is still read by the code. Do not rely on it for new work, but do not be surprised to see it.
- `policy` — a policy trajectory, downscaled and typically compared against a BAU row via `comparison_counterfactual_labels`.

## 7.2 Example

The shipped `standard_scenarios.csv`: a Rwanda baseline, a BAU trajectory, and a policy row that shifts the BAU using regional projections.

```
scenario_label,scenario_type,aoi,exogenous_label,climate_label,model_label,counterfactual_la
baseline_luh2-message,baseline,RWA,baseline,rcp45,luh2-message,,2017,,2017,2017,,add2015,luh
ssp2_rcp45_luh2-message_bau,bau,RWA,ssp2,rcp45,luh2-message,bau,2030 2050,baseline_luh2-mess
ssp2_rcp45_luh2-message_bau_shift,policy,RWA,ssp2,rcp45,luh2-message,bau_shift,2030 2050,bas
```

> **Known wart:** the shipped CSVs currently write `base_year_lulc_path` as `lulc\esa\lulc_esa_2017.tif`, with backslashes. The devstack rule is forward slashes everywhere, on every platform; these should be normalized.

## 7.3 Where the file lives

Put your scenarios CSV in an `input_template/` directory beside your run file. It is tracked in git; ProjectFlow copies anything missing into the project's untracked `input/` on first run and never overwrites the working copy. Name it in your run file's `__main__` block:

```
p.scenario_definitions_filename = 'my_project_scenarios.csv'
```

If the named file is absent from both `input/` and `base_data`, SEALS generates a default one in the project's `input/` so a first run still works — edit that copy, or better, add your own to `input_template/`.

# Part I

# Model Coupling

# 8 MAgPIE

## 8.1 Overview

The Model of Agricultural Production and Its Impact on the Environment (MAgPIE) is a global food and land system modelling framework. It uses a wide range of spatially-explicit biophysical and socioeconomic information to model global land-use dynamics throughout the 21st century. MAgPIE operates at two spatial scales that include a coarse scale of twelve model regions with similar socioeconomic characteristics and a subregional level that combines biophysical information, such as suitability for agricultural production (crop yield potential, irrigation water requirements, travel time to urban markets etc.), to determine the cost-effectiveness of different land-use activities. Both these resolutions, however, are typically too coarse to assess how global land-use dynamics could drive landscape-scale changes and associated changes in biodiversity and ecosystems service supply, which is a key sustainability challenge.

The Spatial Economic Allocation Landscape Simulator (SEALS) model can therefore be applied to spatially allocate projected land-use dynamics to spatial scale that is relevant for assessing landscape and biodiversity change. SEALS uses empirically calibrated adjacency relationships, physical suitability and conversion eligibility information at a spatial resolution of 300 x 300 Meter to allocate land use activities across space.

In the following this tutorial describes how both the MAgPIE and SEALS models can be linked using some of the functionalities provided by MAgPIE and SEALS to processes model outputs (MAgPIE) or inputs (SEALS) respectively. This tutorial therefore assumes some familiarity with the MAgPIE model - in particular how to set up model runs. Basic tutorials on how to run the MAgPIE model can be found here https://magpiemodel.github.io/tutorials/. A full documentation of the MAgPIE model (release version 4.7.2) can be accessed under https://rse.pik-potsdam.de/doc/magpie/4.7.2/ (substitute the version number 4.7.2 with any newer release version number to get the latest documentation).

## 8.2 Getting MAgPIE itself running

Orientation notes for setting MAgPIE up, if you have not run it before. The MAgPIE tutorials linked above are the authoritative source; these are the points worth knowing before you start.

- Clone the development branch:

```
git clone -b develop https://github.com/magpiemodel/magpie.git magpie_develop
```

- The `input` and `output` folders are in `.gitignore`.
- MAgPIE has a config folder. `default.cfg` in it is very large, so rather than editing it directly, use a start script that reads lines from the default and overwrites the ones it needs.
- Start scripts live in the (non-ignored) `start` folder — for example `test_runs.R`.
- To run GAMS, see `start/start_functions.R`.

## 8.3 Create MAgPIE outputs for SEALS by using output scripts

The MAgPIE model provides several output scripts that can be executed after a model run and are stored under `scripts/output` in the MAgPIE folder. Two scripts are particularly relevant for providing MAgPIE outputs to SEALS: the `extra/disaggregation.R` script and the `extra/reportMAgPIE2SEALS.R` script. The `extra/disaggregation.R` script disaggregates coarse-resolution land-use projections to a spatial resolution of 0.5 degrees and is run by default after each model run. The `extra/reportMAgPIE2SEALS.R` script, on the other hand, generates a NetCDF file that can readily be used by SEALS but is *not* run by default.

The output scripts can be selected and executed via the command line. Therefore, on the command prompt, navigate to the MAgPIE model folder and use the following command:

```
Rscript output.R
```

On the command line you are now asked for which of the model runs you would like to do the postprocessing. You can choose `all`, select a specific run, or run the postprocessing for a selection of runs by choosing `search by the pattern.`

```
CLUSTER2015 - vjeetze@login01:/p/projects/magpie/users/vjeetze/magpie/other/f_magpie-seals_tutorial> Rscript output.R
Global .Rprofile loaded! (R version 4.1.2 (2021-11-01))
Hi Pat,

Your .Rprofile was loaded!

Checking for updates... Update check done.
Checking package version requirements... Requirements check done.

Choose runs:
1: all
2: weeklyTests_SSP2EU-NDC/
3: weeklyTests_SSP2EU-PkBudg650/
4: weeklyTests_SSP2EU-Ref/
5: Search by the pattern.
Number:
```

After this you can select the output script that you want to execute. Choose `extra`.

```
Main selection of MAgPIE output scripts
 -------------------------------------------------------------------
 -> Scripts in this selection are actively managed and work out of  <-
 -> the box. Please provide a yml header to your script specifying  <-
 -> "description" and "comparison script", otherwise it may not be  <-
 ->                      processed correctly.                       <-
 -------------------------------------------------------------------
 1:           output check | check output for known problems
 2:             rds report | extract report in rds and mif format from
                           | run
 3:           merge report | Merges report.mif files from several runs
                           | into a single mif file
 4:         rds report iso | extract report in rds format from run
 5:             validation | creates a validation pdf file for a run
                           | (long version - single crop types)
 6:       validation short | creates a validation pdf file for a run
                           | (short version - aggregated crop types)
 7: comparison validation | Creates an validation pdf out of several
                           | runs
 8:          runBlackmagicc | Append MAGICC7 warming pathways to a
                           | MAgPIE run's report.mif
 9:        validation cell | Comparison at cellular level of land and
                           | crop types with LUH2 and MAPSPAM

Alternatively, choose a output script from another selection:
10:                  extra | Additional MAgPIE output scripts
11:               projects | Project-specific MAgPIE output scripts
12:             deprecated | Deprecated scripts
Choose a output script or folder: 10
```

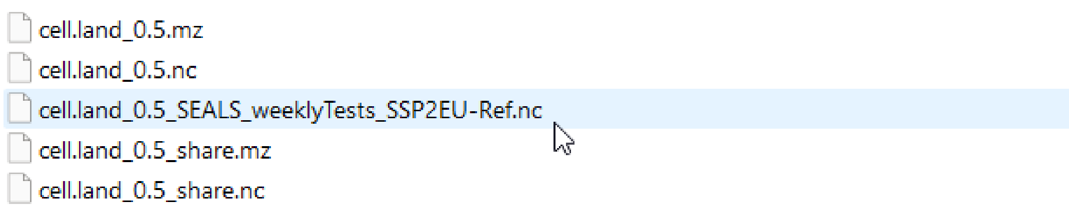
Next select `reportMAgPIE2SEALS`.

```
Additional MAgPIE output scripts
 ---------------------------------------------------------------------
 ->      Scripts in this selection might require some manual        <-
 -> adjustments before they work with the given MAgPIE version as   <-
 ->   they are not necessarily updated along with changes in the    <-
 ->     model code. Please provide a yml header to your script      <-
 -> specifying "description" and "comparison script", otherwise it  <-
 ->                  may not be processed correctly.                <-
 ---------------------------------------------------------------------
 1:                    aff area | extracts afforstation area from
                                | multiple runs and creates a PDF
                                | (useful template for other
                                | variables)
 2:          disaggregation LUH2 | Downscale MAgPIE results to 0.25
                                | degree resolution in LUH2 format for
                                | ISIMIP 3b
 3:               disaggregation | Interpolates land pools to 0.5
                                | degree resolution
 4:                     emulator |
 5:          force runstatistics | Forces runstatistics submission in
                                | central repo
 6:           ForestChangeCluster | Compares Forest Area Change at
                                | Cluster Level
 7:                      highres | starts a run with higher resolution
                                | in parallel mode (each region is
                                | solved individually) using trade
                                | patterns from an existing run
 8:                 land cluster | Write land use information on
                                | cluster level to land_cluster.gpkg
                                | (GeoPackage)
 9:                    modelstat | checks the modelstat of several runs
10:           reportMAgPIE2REMIND | Write only those variables into a
                                | report that are relevant for the
                                | REMIND coupling
11:            reportMAgPIE2SEALS | Modifies gridded MAgPIE land use
                                | patterns so that they can be read in
                                | by the Spatial Economic Allocation
                                | Landscape Simulator (SEALS)
12:                     resubmit | re-submits runs to different queues
                                | (e.g. priority)
13:                      runtime | Checks the runtime of several runs
14:            timestep duration | Makes a png file containing solve
                                | time for each step
15: validation existing report | Creates a validation pdf out of an
                                | existing mif file, speeding up the
                                | process.
Choose a script: 11
```

Finally, choose the run submission type, e.g. `Direct execution`.

```
Choose submission type:
1: SLURM standby
2: SLURM standby maxMem
3: SLURM priority
4: SLURM priority maxMem
5: Direct execution
6: Background execution
7: Debug mode
Number: 5
```

All outputs are written to a subfolder in the `output` folder within the MAgPIE directory. This subfolder (or run folder) is created automatically at the beginning of a model run and its name is a combination of the run title and date (For further information see https://magpiemodel.github.io/tutorials/t06-changingconfig). Before you run `reportMAgPIE2SEALS`, please make sure that the script `extra/disaggregation.R` was executed without error. If the script `extra/disaggregation.R` was completed successfully, the run folder should contain the file `cell.land_0.5_share.mz`. Once the script `extra/reportMAgPIE2SEALS.R` has finished the run folder should contain a file named `cell.land_0.5_SEALS_<run title>.nc`. This file contains spatially-explicit information on the share of total land per grid cell used for different land-use activities (cropland, pasture, forest, urban etc.) projected until 2050, which can now be used as an input for SEALS.




### 8.3.1 Change your start script to generate MAgPIE outputs for SEALS

If you are familiar with setting up a start script for MAgPIE runs (see https://magpiemodel.github.io/tutorials/t07-startscript), an alternative way of generating MAgPIE outputs for SEALS is by adding `"extra/reportMAgPIE2SEALS"` to the setting

```
cfg$output <- c("output_check", "extra/disaggregation", "rds_report")
```

so that it reads

```
cfg$output <- c("output_check", "extra/disaggregation", "rds_report", "extra/reportMAgPIE2SE
```

As the order of the vector determines the order of execution, please make sure that "extra/reportMAgPIE2SEALS" is called after "extra/disaggregation".

## 8.4 Setting up SEALS to downscale MAgPIE outputs

After generating the output cell.land_0.5_SEALS_<run title>.nc, the following describes how this information can be used as input for the SEALS model.

First, we need to relate the land use types of the MAgPIE model to the land cover types used by the SEALS model. While MAgPIE separates seven different main land types (cropland, grassland, primary forest, secondary forest, forestry, non-forest vegetation & urban), the SEALS model only allocates changes for 5 main land types (urban, cropland, grassland, forest & other natural vegetation). Below a mapping between the different land types is shown.

```
src_id,dst_id,src_label,dst_label,src_description,dst_description
6,1,urban,urban,urban land,
1,2,crop,cropland,cropland,
2,3,past,grassland,grassland,
3,4,primforest,forest,primary forest,
4,4,secdforest,forest,secondary forest,
5,4,forestry,forest,forestry,
7,5,other,othernat,non-forest vegetation,
```

Copy and paste this mapping into a CSV file and save it, e.g. under base_data/seals/default_inputs/magpie_s
The folder base_data/seals/default_inputs also contains other mappings such as a mapping betweehn LUH2v2 land types and SEALS. This mapping can now be used to set up the **scenarios CSV**. The scenarios CSV provides SEALS with all necessary information on the input data (or MAgPIE outputs), including the path to the MAgPIE outputs. It also allows the user to set up multiple SEALS runs (e.g. in the case of different scenario runs) in one go. The file is named per project — the one shipped with SEALS is standard_scenarios.csv — and your run file points at it with p.scenario_definitions_filename. One way to better understand it is to run SEALS on the test data without further modification, which generates a default in the project folder that you can then tailor to your needs. The columns are documented in Custom scenarios.

The following shows how an example scenario_definitions_tutorial.csv is created that provides relevant information for a set of different MAgPIE test runs. Assume we have created land use projections for three different test scenarios: a reference 'business-as-usual' scenario (weeklyTests_SSP2EU-Ref), a low ambition mitigation scenario, in which current nationally determined contributions (NDCs) under the Paris agreement for the land sector are fulfilled (weeklyTests_SSP2EU-NDC) and a highly ambitious mitigation scenario in line with the 1.5 degree target (weeklyTests_SSP2EU-PkBudg650). For each of the scenarios, we have created gridded land-use projections in a format that can be read by SEALS using the output script `extra/reportMAgPIE2SEALS.R` (see above).

Here, the `scenario_definitions_tutorial.csv` contains a line for each scenario specified under `scenario_label`. Note that the `Baseline` always needs to be defined as well. In the years column you can specify the years for which SEALS should do a downscaling. Multiple years can be separated by a space.

| | column 1 | column 2 | column 3 | column 4 | column 5 | column 6 | column 7 | column 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | scenario_label | scenario_type | aoi | exogenous_label | climate_label | model_label | counterfactual_label | years |
| 2 | Baseline | baseline | global | baseline | | magpie | | 2020 |
| 3 | weeklyTests_SSP2EU-Ref | bau | global | ssp2 | rcp26 | magpie | weeklyTests_SSP2EU-Ref | 2030 2050 |
| 4 | weeklyTests_SSP2EU-NDC | policy | global | ssp2 | rcp26 | magpie | weeklyTests_SSP2EU-NDC | 2030 2050 |
| 5 | weeklyTests_SSP2EU-PkBudg650 | policy | global | ssp2 | rcp26 | magpie | weeklyTests_SSP2EU-PkBudg650 | 2030 2050 |

Set the path to the gridded MAgPIE outputs in the column `coarse_projections_input_path`

| | column 1 | | column 14 | column 15 |
|---|---|---|---|---|
| 1 | scenario_label | ent | coarse_projections_input_path | lulc_src_label |
| 2 | Baseline | | C:/Users/vjeetze/PIK/MAgPIE/other/magpie_seals_tutorial/cell.land_0.5_SEALS_weeklyTests_SSP2EU-Ref.nc | esa |
| 3 | weeklyTests_SSP2EU-Ref | | C:/Users/vjeetze/PIK/MAgPIE/other/magpie_seals_tutorial/cell.land_0.5_SEALS_weeklyTests_SSP2EU-Ref.nc | esa |
| 4 | weeklyTests_SSP2EU-NDC | | C:/Users/vjeetze/PIK/MAgPIE/other/magpie_seals_tutorial/cell.land_0.5_SEALS_weeklyTests_SSP2EU-NDC.nc | esa |
| 5 | weeklyTests_SSP2EU-PkBudg650 | | C:/Users/vjeetze/PIK/MAgPIE/other/magpie_seals_tutorial/cell.land_0.5_SEALS_weeklyTests_SSP2EU-PkBudg650.nc | esa |

The `lulc_correspondence_path` gives the path to a mapping between ESA CCI land cover classes and the aggregated land cover types used in SEALS.

| | column 1 | column 17 | column 18 | column 19 |
|---|---|---|---|---|
| 1 | scenario_label | lulc_correspondence_path | coarse_src_label | coarse_simplification_label |
| 2 | Baseline | seals/default_inputs/esa_seals7_correspondence_mosaic-natural.csv | magpie | seals7 |
| 3 | weeklyTests_SSP2EU-Ref | seals/default_inputs/esa_seals7_correspondence_mosaic-natural.csv | magpie | seals7 |
| 4 | weeklyTests_SSP2EU-NDC | seals/default_inputs/esa_seals7_correspondence_mosaic-natural.csv | magpie | seals7 |
| 5 | weeklyTests_SSP2EU-PkBudg650 | seals/default_inputs/esa_seals7_correspondence_mosaic-natural.csv | magpie | seals7 |

The default mapping in SEALS is stored under `base_data\seals\default_inputs\esa_seals7_correspondenc` For MAgPIE-SEALS couplings, however, it is recommended to use a mapping that classifies "crop_natural_mosaic" as "othernat" and "natural_crop_mosaic" as "forest" to better align the total cropland extent of the ESA map with the cropland initialisation in MAgPIE. That there is an overestimation of the cropland extent in the ESA CCI data is a well-know issue. Thus, the lines in `esa_seals7_correspondence.csv`

```
6,,30,2,crop_natural_mosaic,cropland
7,,40,2,natural_crop_mosaic,cropland
```

are changed to

```
6,,30,2,crop_natural_mosaic,othernat
7,,40,2,natural_crop_mosaic,forest
```

in `esa_seals7_correspondence_mosaic-natural.csv`.

Next, we set the path to the mapping between MAgPIE land use types to SEALS land cover types, as defined above

| | column 1 | column 20 | column 21 | column 22 |
|---|---|---|---|---|
| 1 | scenario_label | coarse_correspondence_path | lc_class_varname | dimensions |
| 2 | Baseline | seals/default_inputs/magpie_seals7_correspondence.csv | all_variables | time |
| 3 | weeklyTests_SSP2EU-Ref | seals/default_inputs/magpie_seals7_correspondence.csv | all_variables | time |
| 4 | weeklyTests_SSP2EU-NDC | seals/default_inputs/magpie_seals7_correspondence.csv | all_variables | time |
| 5 | weeklyTests_SSP2EU-PkBudg650 | seals/default_inputs/magpie_seals7_correspondence.csv | all_variables | time |

We also specify a set of regressors used during downscaling. Tables of trained model coefficients used by SEALS can be found under `base_data/seals/default_inputs`. Here we can also align some model constraints between MAgPIE and SEALS. If, for example, compliance with currently implemented land policies such as protected areas is assumed in MAgPIE, these constraints also need to be reflected during the downscaling. We here therefore add an additional multiplicative regressor that precludes expansion of agricultural land (cropland and grassland) in legally protected areas as reported by the WDPA database, which is in line with the assumption in MAgPIE.

| | column 1 | column 2 | column 3 | column 4 | column 5 | column 6 | column 7 | column 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | spatial_regressor_name | data_location | type | urban | cropland | grassland | forest | othernat |
| 11 | grassland_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_grassland.tif | additive | 0.005555556 | 0.018888889 | 0 | 0.041666667 | -0.026111111 |
| 12 | forest_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_forest.tif | additive | -0.019444444 | -0.016666667 | -0.002666667 | 0 | 0.033444444 |
| 13 | othernat_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_othernat.tif | additive | 0.01 | 0.144444444 | 0.060111111 | 0.02 | 0 |
| 14 | water_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_water.tif | additive | 0 | 0 | 0 | 0 | 0 |
| 15 | other_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_other.tif | additive | -1.119444444 | 0.001666667 | 0.126666667 | 0.061111111 | -0.023333333 |
| 16 | urban_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_url | additive | 1.713888889 | -1122.233444 | -11.13055556 | 0.041666667 | -1122.241667 |
| 17 | cropland_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_cro | additive | 0.105555556 | 0.333444444 | 0.022222222 | 0 | -11.24444444 |
| 18 | grassland_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_gra | additive | 0.054444444 | 0.005444444 | 0.38 | 0.018222222 | 0.085555556 |
| 19 | forest_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_fo | additive | -0.122222222 | 1111.065556 | -0.011 | 1.555 | -0.022233333 |
| 20 | othernat_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_otl | additive | 0.010888889 | 0 | 0.019444444 | -0.122222222 | 0.466666667 |
| 21 | water_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_wa | additive | 0.036555556 | -112.2027778 | -112.24 | -1.105555556 | -1133.322333 |
| 22 | other_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_otl | additive | -0.127777778 | -112.2555556 | -1112.144444 | -1111.133333 | 0.005555556 |
| 23 | urban_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_url | additive | -0.072222222 | -11.52222222 | -111.2638889 | -0.093333333 | -0.087777778 |
| 24 | cropland_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_cro | additive | 0.068888889 | 0.162333333 | -0.016666667 | 0.122222222 | 0.065555556 |
| 25 | grassland_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_gra | additive | 0.100222222 | -0.025 | 0.431111111 | -0.026677778 | -0.041666667 |
| 26 | forest_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_fo | additive | 0.133222222 | 0.367777778 | 0.076333333 | 0.68888 | 0.113333333 |
| 27 | othernat_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_otl | additive | 0 | -0.073111111 | 0.024444444 | -0.005555556 | 0.152777778 |
| 28 | water_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_wa | additive | 0.091666667 | 0.005 | -1111.105444 | -1.092777778 | -0.002222222 |
| 29 | other_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_otl | additive | 0.045555556 | 0.15 | -1111.077778 | -0.008333333 | -110.89 |
| 30 | soil_organic_content_1m_30s | seals/static_regressors/soil_organic_content.tif | additive | 0.027777778 | -0.15 | 110.9777778 | -111.14 | -0.027777778 |
| 31 | bio_12 | seals/static_regressors/precip_mm.tif | additive | 11.11944444 | -0.994444444 | 1.14 | -1.075 | 11.00444444 |
| 32 | alt | seals/static_regressors/alt_m.tif | additive | -0.104444444 | 0.085 | -0.024888889 | -0.037788889 | 0.01 |
| 33 | bio_1 | seals/static_regressors/temperature_c.tif | additive | -0.022111111 | 0.044444444 | -0.011111111 | -0.01 | -0.001111111 |
| 34 | minutes_to_market_30s | seals/static_regressors/travel_time_to_market_mins.tif | additive | 0.016122222 | 0.21 | 0.005555556 | 10.555 | -0.034333333 |
| 35 | pop_30s | seals/static_regressors/pop.tif | additive | 0 | 0 | 0 | 0 | 0 |
| 36 | CEC_1m_30s | seals/static_regressors/soil_cec.tif | additive | 0 | -0.016666667 | 0 | 111.0722222 | 0 |
| 37 | clay_percent_1m_30s | seals/static_regressors/clay_percent.tif | additive | -0.051111111 | 0.02 | -0.186111111 | -0.046222222 | 1.084333333 |
| 38 | ph_1m_30s | seals/static_regressors/ph.tif | additive | 0 | 0.1 | 0 | 0 | 0 |
| 39 | sand_percent_1m_30s | seals/static_regressors/sand_percent.tif | additive | 0.034444444 | 0.018333333 | -0.037777778 | -0.048888889 | -0.001111111 |
| 40 | silt_percent_1m_30s | seals/static_regressors/silt_percent.tif | additive | -0.012777778 | -0.165 | 0 | -0.059011111 | -0.146111111 |
| 41 | wdpa | seals/static_regressors/magpie_WDPA.tif | multiplicative | 0 | 0 | 0 | 1 | 1 |
| 42 | wetlands_binary | seals/static_regressors/wetlands_binary.tif | additive | 0 | 0 | 0 | 0 | 0 |
| 43 | carbon_above_ground_mg_per_ha_global | seals/static_regressors/carbon_above_ground_mg_per_ha_global.tif | additive | 0 | 0 | 0 | 0 | 0 |

Similarly, the same functionality can also be used to specify policy scenarios, such as the expansion of protected areas in line with the 30x30 goal of the Global Biodiversity Framework. The `seals/static_regressors/magpie_30by30.tif` contains both currently protected areas from the WDPA database and areas that might be protected under a global 30x30 protection scenario.

| | column 1 | column 2 | column 3 | column 4 | column 5 | column 6 | column 7 | column 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | spatial_regressor_name | data_location | type | urban | cropland | grassland | forest | othernat |
| 11 | grassland_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_grassland.tif | additive | 0.005555556 | 0.018888889 | 0 | 0.041666667 | -0.026111111 |
| 12 | forest_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_forest.tif | additive | -0.019444444 | -0.016666667 | -0.002666667 | 0 | 0.033444444 |
| 13 | othernat_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_othernat.tif | additive | 0.01 | 0.144444444 | 0.060111111 | 0.02 | 0 |
| 14 | water_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_water.tif | additive | 0 | 0 | 0 | 0 | 0 |
| 15 | other_binary | lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_other.tif | additive | -1.119444444 | 0.001666667 | 0.126666667 | 0.061111111 | -0.023333333 |
| 16 | urban_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_urk | additive | 1.713888889 | -1122.233444 | -11.13055556 | 0.041666667 | -1122.241667 |
| 17 | cropland_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_cro | additive | 0.105555556 | 0.333444444 | 0.022222222 | 0 | -11.24444444 |
| 18 | grassland_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_gra | additive | 0.054444444 | 0.005444444 | 0.38 | 0.018222222 | 0.085555556 |
| 19 | forest_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_for | additive | -0.122222222 | 1111.065556 | -0.011 | 1.555 | -0.022233333 |
| 20 | othernat_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_oth | additive | 0.010888889 | 0 | 0.019444444 | -0.122222222 | 0.466666667 |
| 21 | water_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_wa | additive | 0.036555556 | -112.2027778 | -112.24 | -1.105555556 | -1133.322333 |
| 22 | other_gaussian_1 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_oth | additive | -0.127777778 | -112.2555556 | -1112.144444 | -1111.133333 | 0.005555556 |
| 23 | urban_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_urk | additive | -0.072222222 | -11.52222222 | -111.2638889 | -0.093333333 | -0.087777778 |
| 24 | cropland_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_cro | additive | 0.068888889 | 0.162333333 | -0.016666667 | 0.122222222 | 0.065555556 |
| 25 | grassland_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_gra | additive | 0.100222222 | -0.025 | 0.431111111 | -0.026677778 | -0.041666667 |
| 26 | forest_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_for | additive | 0.133222222 | 0.367777778 | 0.076333333 | 0.68888 | 0.113333333 |
| 27 | othernat_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_oth | additive | 0 | -0.073111111 | 0.024444444 | -0.005555556 | 0.152777778 |
| 28 | water_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_wa | additive | 0.091666667 | 0.005 | -1111.105444 | -1.092777778 | -0.002222222 |
| 29 | other_gaussian_5 | lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_oth | additive | 0.045555556 | 0.15 | -1111.077778 | -0.008333333 | -110.89 |
| 30 | soil_organic_content_1m_30s | seals/static_regressors/soil_organic_content.tif | additive | 0.027777778 | -0.15 | 110.9777778 | -111.14 | -0.027777778 |
| 31 | bio_12 | seals/static_regressors/precip_mm.tif | additive | 11.11944444 | -0.994444444 | 1.14 | -1.075 | 11.00444444 |
| 32 | alt | seals/static_regressors/alt_m.tif | additive | -0.104444444 | 0.085 | -0.024888889 | -0.037788889 | 0.01 |
| 33 | bio_1 | seals/static_regressors/temperature_c.tif | additive | -0.022111111 | 0.044444444 | -0.011111111 | -0.01 | -0.001111111 |
| 34 | minutes_to_market_30s | seals/static_regressors/travel_time_to_market_mins.tif | additive | 0.016122222 | 0.21 | 0.005555556 | 10.555 | -0.034333333 |
| 35 | pop_30s | seals/static_regressors/pop.tif | additive | 0 | 0 | 0 | 0 | 0 |
| 36 | CEC_1m_30s | seals/static_regressors/soil_cec.tif | additive | 0 | -0.016666667 | 0 | 111.0722222 | 0 |
| 37 | clay_percent_1m_30s | seals/static_regressors/clay_percent.tif | additive | -0.051111111 | 0.02 | -0.186111111 | -0.046222222 | 1.084333333 |
| 38 | ph_1m_30s | seals/static_regressors/ph.tif | additive | 0 | 0.1 | 0 | 0 | 0 |
| 39 | sand_percent_1m_30s | seals/static_regressors/sand_percent.tif | additive | 0.034444444 | 0.018333333 | -0.037777778 | -0.048888889 | -0.001111111 |
| 40 | silt_percent_1m_30s | seals/static_regressors/silt_percent.tif | additive | -0.012777778 | -0.165 | 0 | -0.059011111 | -0.146111111 |
| 41 | 30by30 | seals/static_regressors/magpie_30by30.tif | multiplicative | 0 | 0 | 0 | 1 | 1 |
| 42 | wetlands_binary | seals/static_regressors/wetlands_binary.tif | additive | 0 | 0 | 0 | 0 | 0 |
| 43 | carbon_above_ground_mg_per_ha_global | seals/static_regressors/carbon_above_ground_mg_per_ha_global.tif | additive | 0 | 0 | 0 | 0 | 0 |

Finally, the path to the regressor used during the downscaling must also be added to the scenarios CSV, while the base land-cover data `base_year_lulc_path` must refer to the same year as defined in the `Baseline` case.

| | column 1 | column 20 | column 21 | column 22 | column 23 | column 24 |
|---|---|---|---|---|---|---|
| 1 | scenario_label | espondence_path | lc_class_varname | dimensions | calibration_parameters_source | base_year_lulc_path |
| 2 | Baseline | lt_inputs/magpie_seals7_correspondence.csv | all_variables | time | seals/default_inputs/magpie_global_coefficients_wdpa_base.csv | lulc/esa/lulc_esa_2020.tif |
| 3 | weeklyTests_SSP2EU-Ref | lt_inputs/magpie_seals7_correspondence.csv | all_variables | time | seals/default_inputs/magpie_global_coefficients_wdpa_base.csv | lulc/esa/lulc_esa_2020.tif |
| 4 | weeklyTests_SSP2EU-NDC | lt_inputs/magpie_seals7_correspondence.csv | all_variables | time | seals/default_inputs/magpie_global_coefficients_wdpa_base.csv | lulc/esa/lulc_esa_2020.tif |
| 5 | weeklyTests_SSP2EU-PkBudg650 | lt_inputs/magpie_seals7_correspondence.csv | all_variables | time | seals/default_inputs/magpie_global_coefficients_wdpa_base.csv | lulc/esa/lulc_esa_2020.tif |

Save `scenario_definitions_tutorial.csv` in the `input_template/` directory beside your run file — it is tracked in git, and ProjectFlow copies it into the project's `input/` on first run. Then name it in the run file's `__main__` block with `p.scenario_definitions_filename = 'scenario_definitions_tutorial.csv'` and you are good to go.

```
        sys.path.insert(0, custom_seals_path)

    # SEALS will run based on the scenarios defined in a scenario_definitions.csv
    # If you have not run SEALS before, SEALS will generate it in your project's input_dir.
    # A useful way to get started is to to run SEALS on the test data without modification
    # and then edit the scenario_definitions.csv to your project needs.
    # Some of the other test files use different scenario definition csvs
    # to illustrate the technique. If you point to one of these
    # (or any one CSV that already exists), SEALS will not generate a new one.
    # The avalable example files in the default_inputs include:
    # - test_three_scenario_defininitions.csv
    # - test_scenario_defininitions_multi_coeffs.csvs

    p.scenario_definitions_path = os.path.join(p.input_dir, 'scenario_definitions_tutorial.csv')

    # Set defaults and generate the scenario definitions csv if it doesn't exist
```

# Part II

# Worked examples

# 9 Worked example: Kenya DEMETRA-SEALS-InVEST

Workflow Walkthrough

> **Historical report — not a current guide**
>
> This documents the DEMETRA-SEALS-InVEST workflow **as it stood in July 2026**. It is kept as a record of that work and is not maintained against the current API. The run file it describes, `run_ken_demetra.py`, and the `ecosystem_services.py` module it used have since been retired to `seals/seals/old_run_files/` and are no longer supported. Run files have also moved to the `run_project(p)` convention since this was written, so the code shown here will not match what is in the repo.
> For a current walkthrough see Quickstart and Walkthrough.

## 9.1 Overview

This walkthrough guides you through installing, configuring, and running the DEMETRA-SEALS-InVEST workflow. SEALS (Spatial Economic Allocation Landscape Simulator) integrates with InVEST (Integrated Valuation of Ecosystem Services and Tradeoffs) to model land-use change scenarios and their ecosystem service impacts.

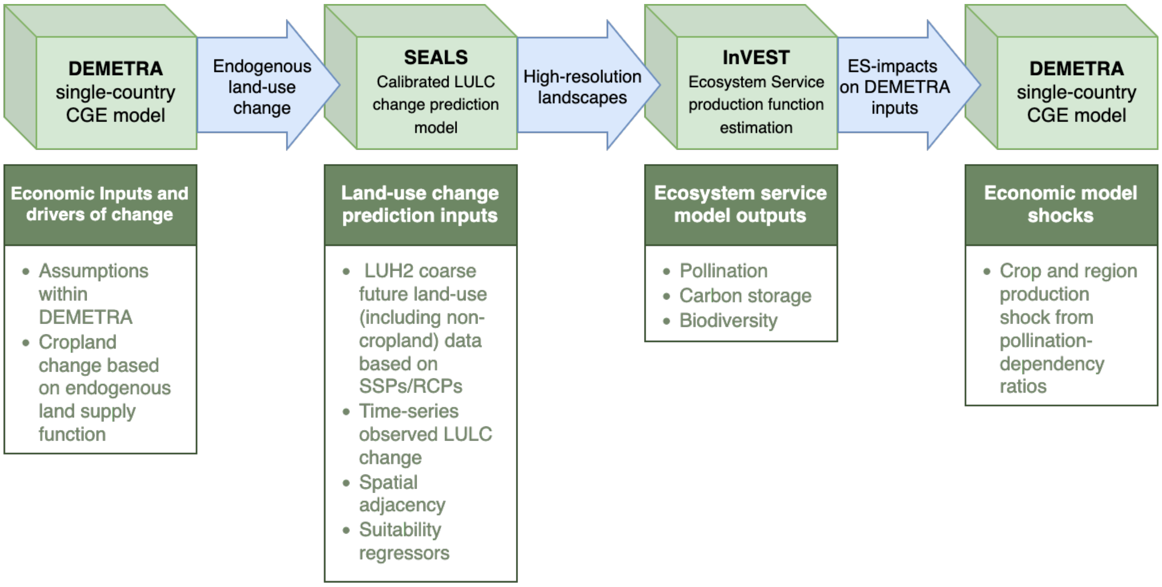



Figure 9.1: Workflow Diagram

---

## 9.2 Installation

The steps below are the SEALS install applied to this example. For the canonical version, see Installation; for the devstack-wide options, see the devstack installation guide.

> **Prerequisites**
>
> Before beginning, ensure you have:
>
> - **Git** - Download here
> - **Miniforge3** (Python distribution) - Download here

> **Basic Hazelbean Installation**
>
> Hazelbean is the core dependency for running SEALS. Install it with these commands:

```
# Initialize conda (may not be needed depending on your shell configuration)
conda init

# Create a new environment (pick your own name in place of <your_env>)
# No Python version: conda picks one hazelbean is built for in the next step
conda create -n <your_env>

# Activate the environment
conda activate <your_env>

# Install hazelbean and the geospatial stack from conda-forge
mamba install hazelbean cython libgdal-hdf5
```

**Note:** The `mamba install` command may take 5-10 minutes to complete.

---

### 9.2.1 SEALS Repository Installation

Step 1: Install C/C++ Compiler

**Windows Users - Choose One Option:**

- **Option 1:** Download from Visual Studio Build Tools and select "Download Build Tools"
- **Option 2:** Use winget command:

  ```
  winget install Microsoft.VisualStudio.2022.BuildTools --force --override "--passive --
  ```

- **Option 3:** Run the `install.bat` file from the Earth Economy Devstack repository root (executes the winget command above)

**Mac/Linux Users:**
Most users will already have Xcode installed. If not:

```
xcode-select --install
```

This installs Xcode Command Line Tools including gcc and clang compilers.

Step 2: Clone SEALS Repository

Clone the repository to your local machine:

```
git clone https://github.com/jandrewjohnson/seals
```

**Recommended location:** `C:/Users/<username>/Files/seals/seals` (Windows) or `/Users/<username>/Files/seals/seals` (Mac/Linux)

Step 3: Install as Editable Package

```
# Activate your conda environment
conda activate <your_env>

# Navigate to the cloned repository
cd C:/Users/<username>/Files/seals/seals

# Install as editable package
pip install -e . --no-deps
```

---

## 9.3 Data Setup

### 9.3.1 File Organization Structure

SEALS uses a specific directory structure to locate data automatically:

#### 9.3.1.1 Base Data (Global Shared Data)

SEALS automatically downloads base data on first run to:

```
<home>/Files/base_data/seals/
<home>/Files/base_data/lulc/
```

#### 9.3.1.2 Additional InVEST data

```
<home>/Files/base_data/global_invest/
<home>/Files/base_data/crops/
```

#### 9.3.1.3 Project Data (Country/Region-Specific)

#### 9.3.1.4 Additional country data

```
<home>/Files/seals/projects/run_ken_demetra/
```

This includes:

- Region border files (shapefiles and raster GeoTIFFs)
- DEMETRA shock files (CSV)
- Downscaling coefficient files (CSV and global rasters)
- Scenario configuration files

---

## 9.4 Testing SEALS

### 9.4.1 Standard Test Run

This basic test verifies your installation is working correctly.

**Steps:**

1. Locate `run_seals.py` in `seals/seals/`
2. Ensure your conda environment is activated:

   ```
   conda activate <your_env>
   ```

3. Run the script:

   ```
   python run_seals.py
   ```

4. Check for output in:

   ```
   <home>/Files/seals/projects/standard_<timestamp>/intermediate/visualization/lulc_pngs/
   ```

**Expected Output:** You should see land-use/land-cover (LULC) maps like:

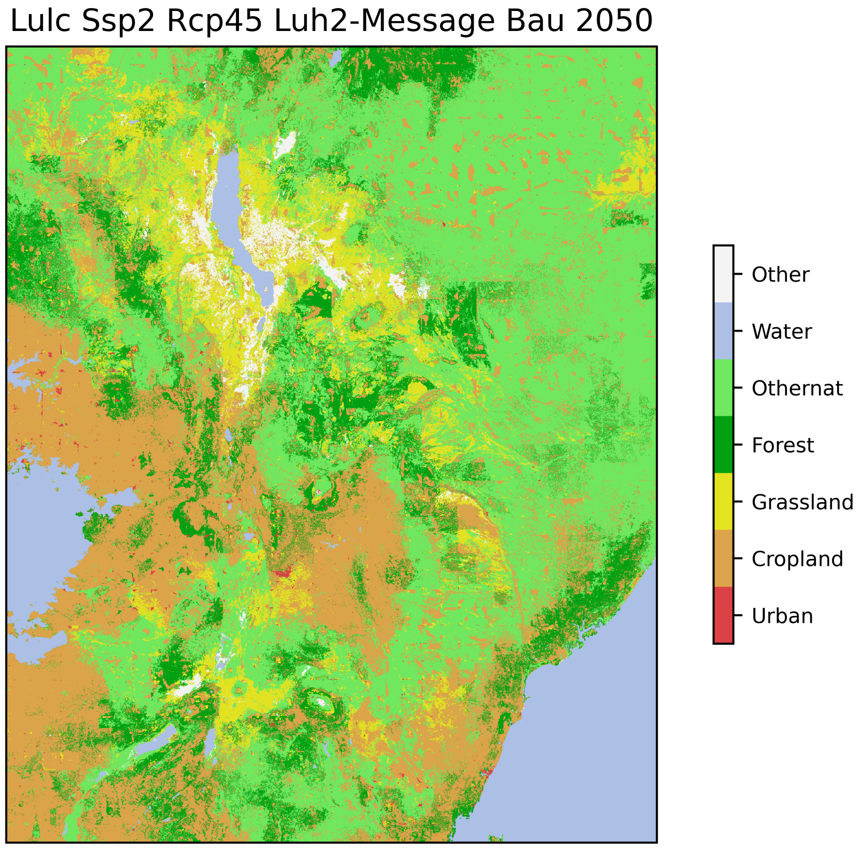



Figure 9.2: Example Output

### 9.4.2 Common Issues

*Problem*: `ValueError: Could not open...` error on first run

*Solution*: The base_data download likely didn't complete. Run the script again now that base_data is fully downloaded.

---

## 9.5 Kenya DEMETRA Scenario

This scenario demonstrates land-use change modeling for Kenya using DEMETRA economic shocks.

### 9.5.1 Key Input Files

**Scripts:**

- Main execution script: `run_ken_demetra.py`

**Data Files:**

| File | File Type | Location |
|---|---|---|
| Scenario Configuration | `ken_scenarios_full.csv` | `projects/run_ken_demetra/` |
| Default Downscaling Coefficients | CSV files + Global rasters | `base_data/seals/static_regressors` `base_data/seals/default_inputs` |
| DEMETRA Shocks | CSV files | `projects/run_ken_demetra/inputs/demetra` |
| Regional Borders | Shapefiles + Raster GeoTIFFs | `projects/run_ken_demetra/inputs/borders` |
| Project Downscaling Coefficients | CSV files + Global rasters | `projects/run_ken_demetra/inputs/static_` `projects/run_ken_demetra/inputs/coeffic` |

**Note:** The structure is flexible, but ensure that the paths defined in the scenario CSV and coefficients CSV are properly defined.

### 9.5.2 Running the Kenya Scenario

```
conda activate <your_env>
cd seals/seals/
python run_ken_demetra.py
```

---

## 9.6 Understanding the Task Tree

When you run `run_ken_demetra.py`, SEALS creates a ProjectFlow object that executes a series of tasks in a hierarchical tree structure. Each task creates outputs in the `intermediate/` directory.

### 9.6.1 Task Hierarchy

For reference, here's the complete task tree as shown in the console output:

```
execute
  project_aoi
  fine_processed_inputs
     generated_kernels
     lulc_clip
     lulc_simplifications
     lulc_binaries
     lulc_convolutions
  coarse_change
     coarse_extraction
     coarse_simplified_proportion
     coarse_simplified_ha
     coarse_simplified_ha_difference_from_previous_year
  regional_change
  allocations
     allocation_zones
         allocation
  stitched_lulc_simplified_scenarios
  visualization
     lulc_pngs
  reproject_lulc_rasters_to_equal_area
  run_invest_carbon
  run_invest_pollination
  calculate_crop_value_and_shock
  calculate_biodiversity_index
  summarize_and_visualize_multi_vector
```

Each task is explained in detail below.

---

## 9.7 Task-by-Task Walkthrough

### 9.7.1 1. Project AOI (Area of Interest)

**Significance:** Defines the geographic boundary for the analysis and creates resolution pyramids for efficient processing at multiple scales.

**What it does:**

1. Loads the area of interest boundary for Kenya from the input shapefile
2. Creates a vector geopackage (`.gpkg`) file for the AOI
3. Generates raster "pyramids" at different resolutions:
    - Fine resolution: Matches the base LULC data (typically 300m)
    - Coarse resolution: Matches the coarse projection data (typically 0.25°)

**Input files:**

- Region shapefile from `projects/run_ken_demetra/inputs/borders`

**Output files:**

```
intermediate/project_aoi/
  aoi_KEN.gpkg                               # Vector boundary
  pyramids/
      aoi_ha_per_cell_coarse.tif            # Coarse resolution (hectares per cell)
      aoi_ha_per_cell_fine.tif              # Fine resolution (hectares per cell)
```

**Significance:**

- The AOI defines the study region
    - Can use default global country shapefile or project-specific
- Pyramids allow multi-scale analysis
- Fine resolution preserves local detail
- Coarse resolution matches global projections

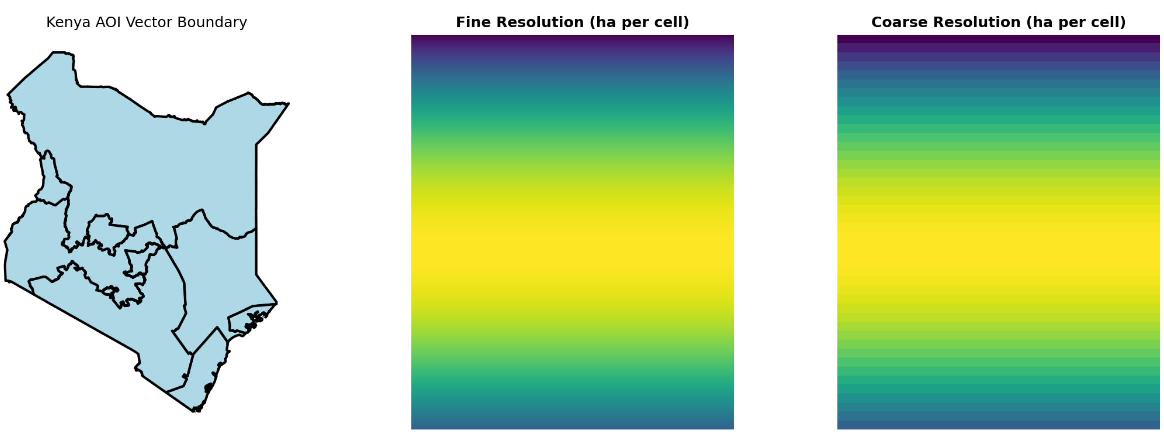



Figure 9.3: Area of Interest for Kenya with resolution pyramids

---

### 9.7.2 2. Fine Processed Inputs

**Significance:** Prepares the baseline land-use/land-cover (LULC) data for allocation by simplifying classifications, creating binary masks, and generating spatial convolutions.

**What it does:**

1. Generated Kernels: Creates Gaussian kernels for spatial smoothing
2. LULC Clip: Clips global LULC data to the Kenya AOI
3. LULC Simplifications: Reclassifies detailed ESA categories into 7 SEALS classes:
   - Cropland
   - Forest
   - Grassland
   - Urban
   - Water
   - Other natural (othernat)
   - Other
4. LULC Binaries: Creates separate binary masks for each class (1 = present, 0 = absent)
5. LULC Convolutions: Applies Gaussian smoothing to capture neighborhood characteristics

**Key Output Structure:**

```
intermediate/fine_processed_inputs/
  generated_kernels/
     gaussian_1.tif                    # Small neighborhood kernel
     gaussian_5.tif                    # Large neighborhood kernel
  lulc/esa/
     lulc_esa_2020.tif                 # Original ESA LULC
     seals7/
         lulc_esa_seals7_2020.tif      # Simplified 7-class LULC
         binaries/2020/                 # Binary masks by class
            binary_esa_seals7_2020_cropland.tif
            binary_esa_seals7_2020_forest.tif
            ...
         convolutions/2020/             # Smoothed neighborhood data
             convolution_esa_seals7_2020_cropland_gaussian_1.tif
             convolution_esa_seals7_2020_cropland_gaussian_5.tif
             ...
```

**Significance:**

- Simplification reduces complexity while preserving key land-use patterns
- Binary masks enable class-specific analysis
- Convolutions capture spatial context
- These features inform the allocation model's decision-making

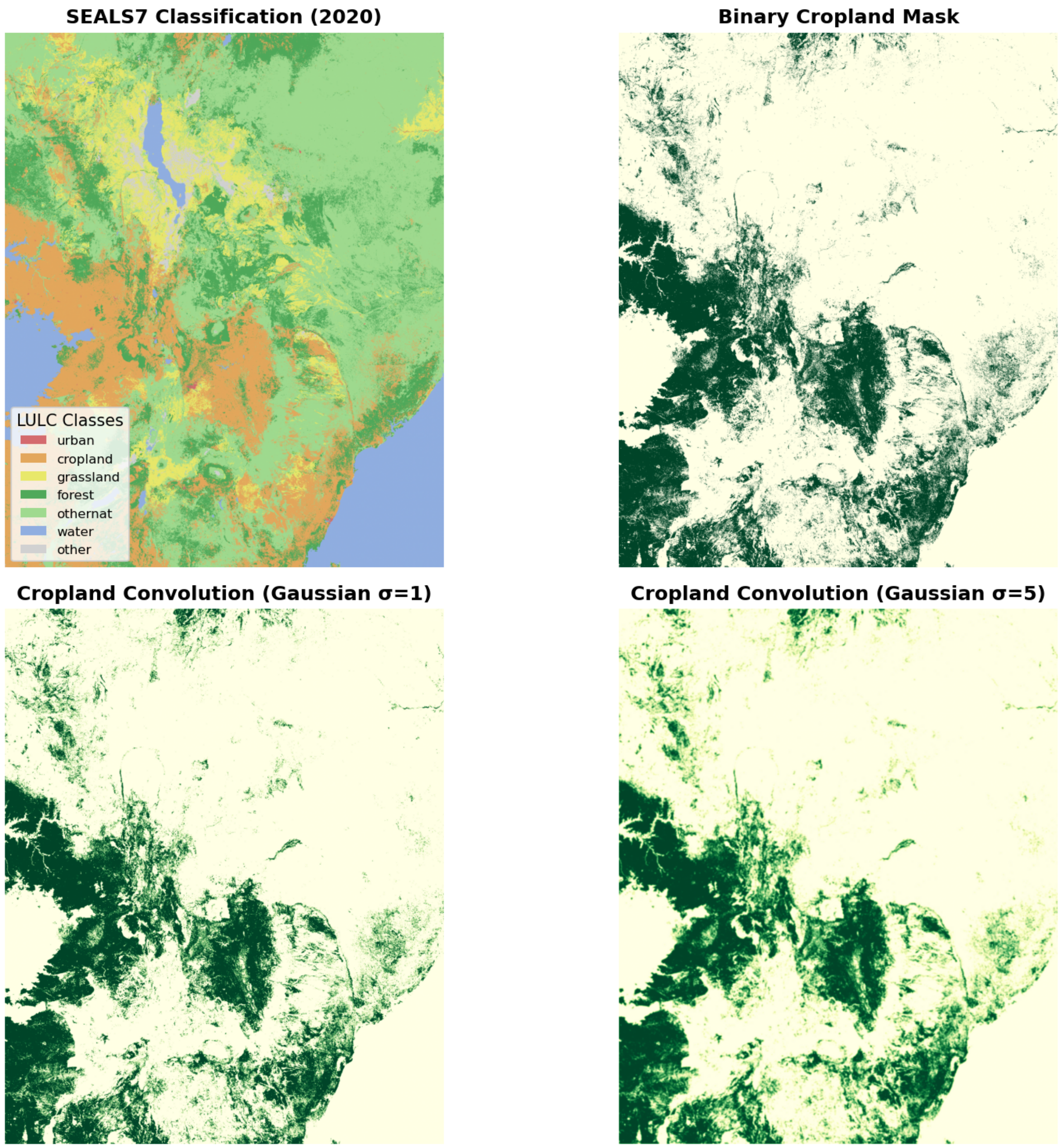



Figure 9.4: Fine processed inputs example: Cropland processing pipeline

---

### 9.7.3 3. Coarse Change

**Significance:** Extracts and processes land-use change projections from global coarse-resolution models (e.g., LUH2) for the Kenya region.

**What it does:**

1. Coarse Extraction: Clips global projection data to Kenya's bounding box and selected year
2. Coarse Simplified Proportion: Calculates proportions of each SEALS7 class
3. Coarse Simplified Ha: Converts proportions to hectares
4. Ha Difference: Calculates change in hectares between time periods (e.g., 2035 → 2050)

**Key Output Example:**

```
intermediate/coarse_change/coarse_simplified_ha_difference_from_previous_year/
  ssp2/rcp45/luh2-message/bau/2050/
      cropland_2050_2035_ha_diff_ssp2_rcp45_luh2-message_bau.tif
      forest_2050_2035_ha_diff_ssp2_rcp45_luh2-message_bau.tif
      ...
```

**Significance:**

- Links exogenous global scenarios (SSP2, RCP4.5) to local allocation
- Tracks changes over time (2035 baseline → 2050 projection)

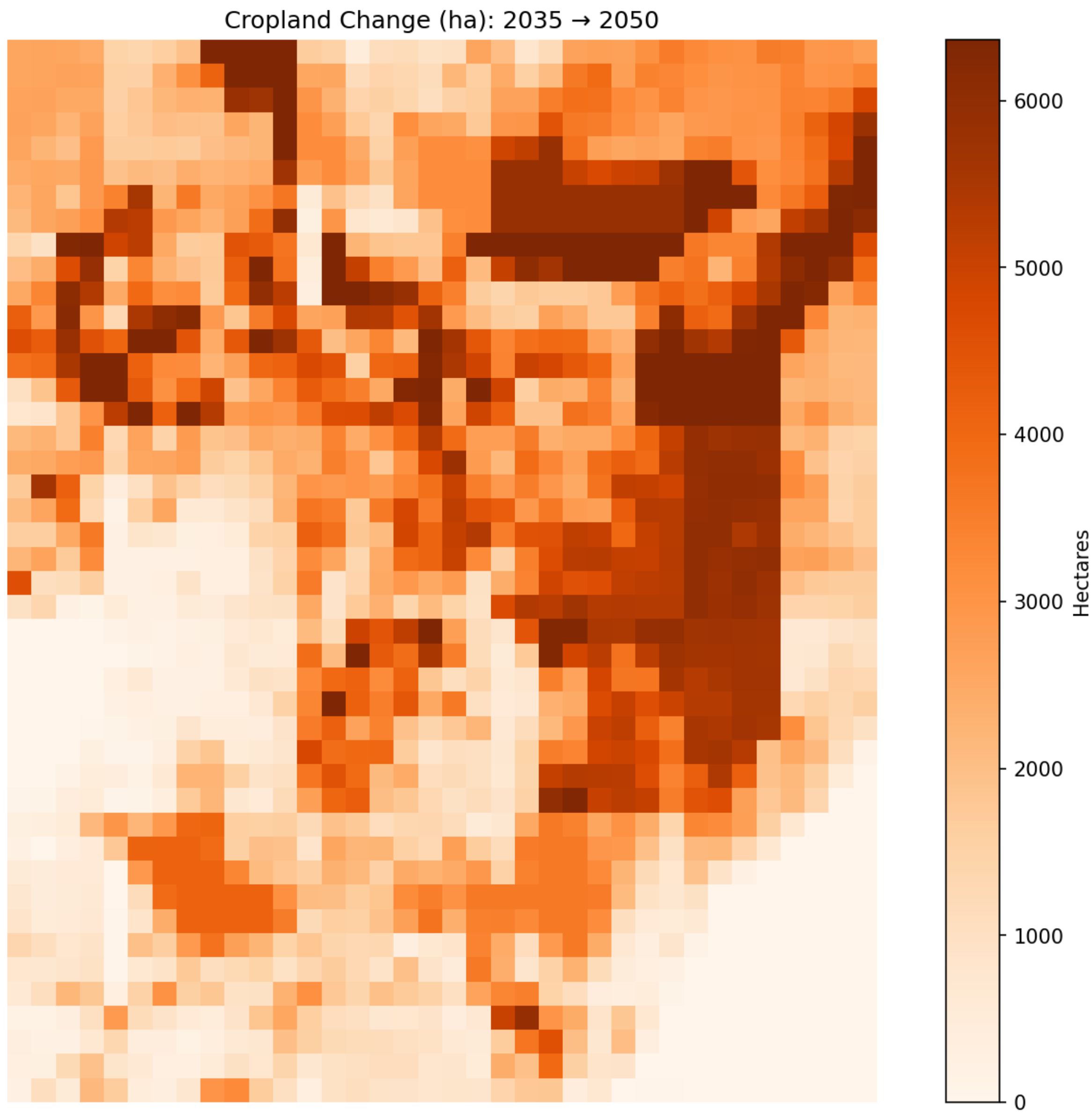



Figure 9.5: Coarse-resolution cropland change (2035-2050) under LP+FP scenario

---

### 9.7.4 4. Regional Change

**Significance:** Adjusts coarse-resolution changes using regional covariates to account for local constraints and opportunities.

**What it does:**

1. Creates a region ID map dividing AOI into separately shocked zones
2. Applies various algorithms (e.g., proportional, covariate sum shift) to downscale from regional to coarse

**Key Outputs:**

```
intermediate/regional_change/
  region_ids.tif                              # Regional zone identifiers
  ssp2/rcp45/luh2-message/LP/2050/
      cropland_2050_2035_ha_diff_ssp2_rcp45_luh2-message_LP_covariate_sum_shift.tif
```

**Significance:**

- Accounts for local factors (infrastructure, topography, current land use)
- Bridges the gap between coarse change and regional change

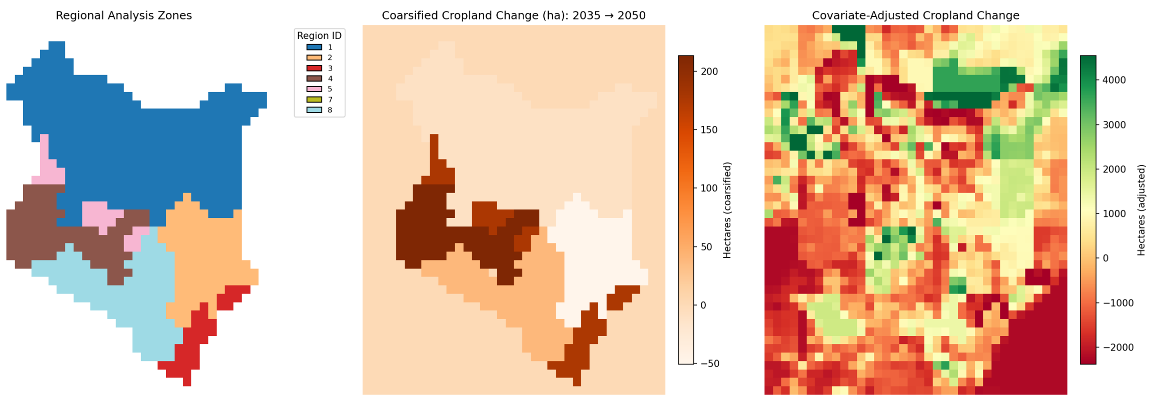



Figure 9.6: Regional + Coarse change for cropland allocation

---

### 9.7.5 5. Allocations

**Significance:** Performs the core SEALS allocation, distributing coarse-resolution changes to fine-resolution pixels based on suitability.

**What it does:**

1. Divides Kenya into allocation zones (e.g., zone 213_84)
2. For each zone, runs an allocation algorithm that:

    - Considers spatial covariates (convolutions, distance, etc.)
    - Respects constraints (protected areas, water bodies)
    - Tracks cumulative change
3. Produces new LULC maps for each scenario and year

**Key Output Structure:**

```
intermediate/allocations/ssp2/rcp45/luh2-message/bau/2050/
  allocation_zones/213_84/allocation/
      block_ha_per_cell_coarse.tif              # Coarse resolution for this block
      cumulative_change_happened.tif            # Tracking allocated changes
      lulc_esa_seals7_ssp2_rcp45_luh2-message_bau_2050.tif  # New LULC
```

**Significance:**

- This is the core of SEALS, where changes actually get placed on the landscape
- Allocation uses machine learning and optimization to find the "best" locations
- Results are spatially explicit: you can see exactly where cropland expanded

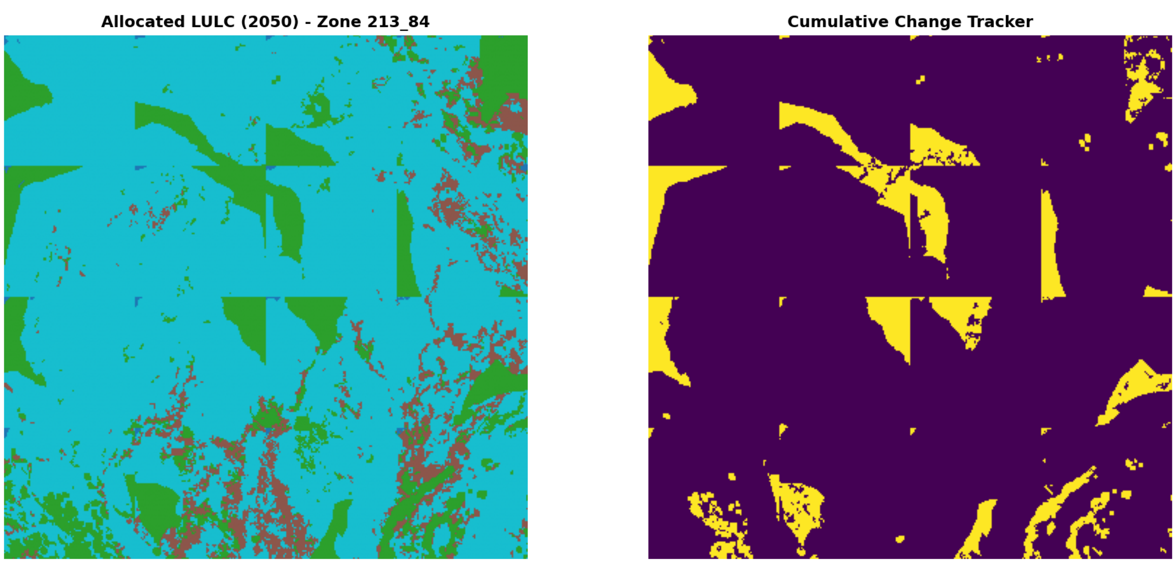



Figure 9.7: Example allocation zone showing LULC projection and cumulative change

---

### 9.7.6 6. Stitched LULC Scenarios

**Significance:** Combines all allocation zones back into complete Kenya-wide LULC maps for each scenario.

**What it does:**

1. Stitches together individual allocation blocks
2. Clips to exact Kenya boundary
3. Generates outputs for all scenario combinations (bau, enablers, es_index, LP, FP, LP+FP)

**Key Outputs:**

```
intermediate/stitched_lulc_simplified_scenarios/
  lulc_esa_seals7_ssp2_rcp45_luh2-message_bau_2050.tif
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP_2050.tif
  lulc_esa_seals7_ssp2_rcp45_luh2-message_FP_2050.tif
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050.tif
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050_clipped.tif
```

**Significance:**

- Provides complete LULC projections
- Enables comparison across scenarios
- Serves as input to ecosystem service models (InVEST)

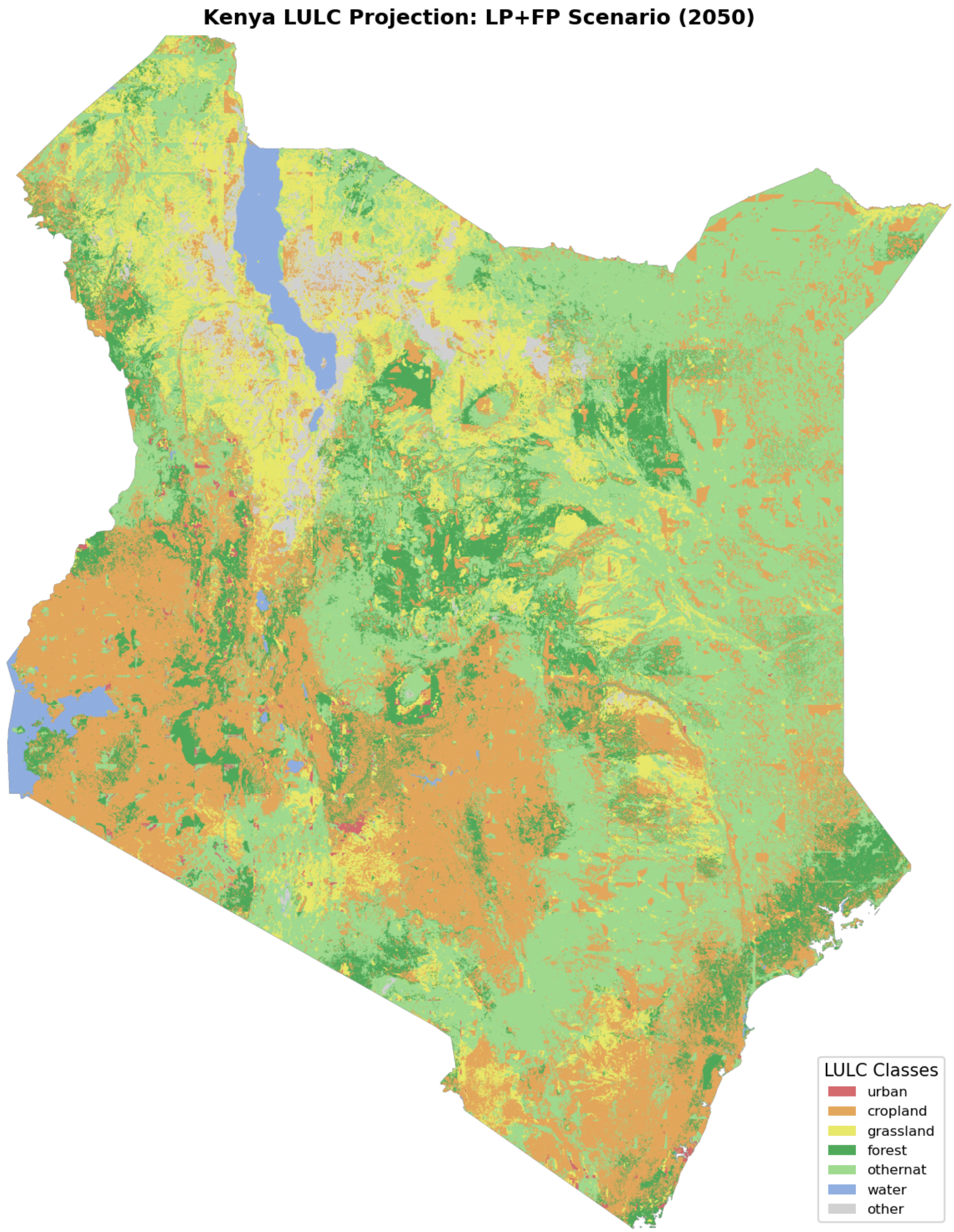



Figure 9.8: Complete LULC projection for Kenya under LP+FP scenario (2050)

---

### 9.7.7 7. Visualization

**Significance:** Creates publication-quality PNG images of LULC projections for easy viewing and sharing.

**What it does:**

1. Renders each stitched LULC raster as a PNG
2. Applies consistent color schemes
3. Adds appropriate labels and legends

**Key Outputs:**

```
intermediate/visualization/lulc_pngs/
  lulc_esa_seals7_ssp2_rcp45_luh2-message_bau_2050.png
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP_2050.png
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050.png
```

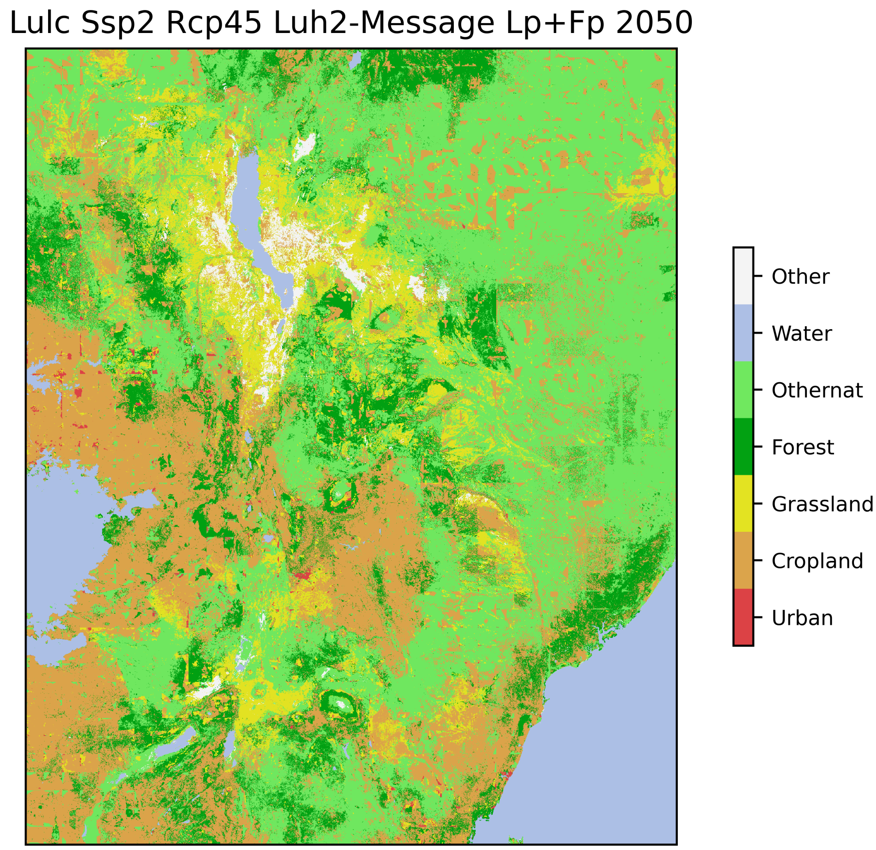



Figure 9.9: Default PNG image for LP+FP scenario

---

### 9.7.8 8. InVEST Modules

**Significance:** Calculates ecosystem service values using Natural Capital Project's InVEST models.

#### 9.7.8.1 8.1 Reproject to Equal Area

Converts LULC maps to equal-area projection (required for accurate area-based calculations).

```
intermediate/reproject_lulc_rasters_to_equal_area/
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050.tif
```

#### 9.7.8.2 8.2 Carbon Storage

Estimates above- and below-ground carbon stocks using the InVEST Carbon model.

**Key Output:**

```
intermediate/run_invest_carbon/
  esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050/
      c_storage_bas.tif                        # Carbon storage (Mg C/ha)
```

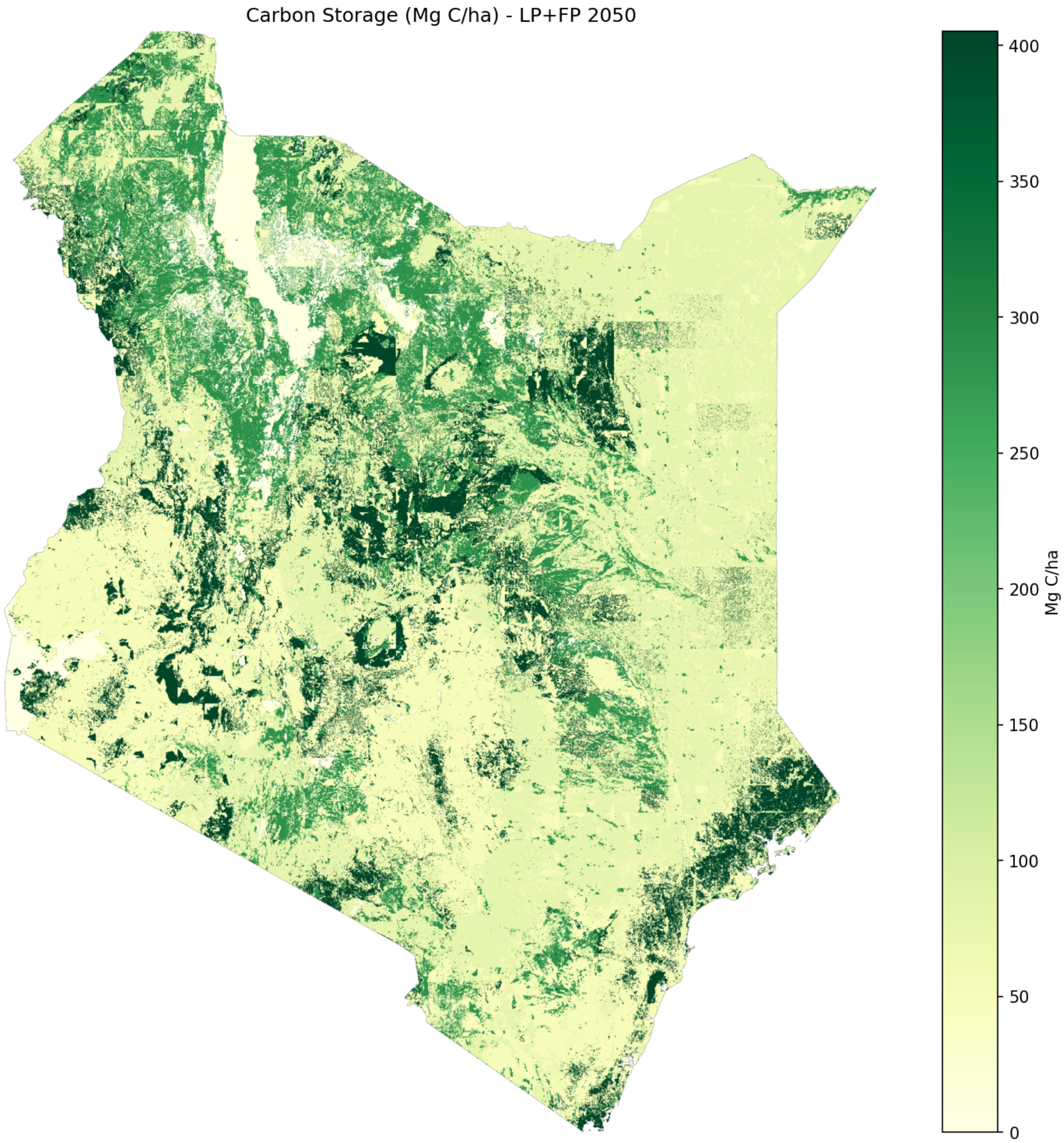



Figure 9.10: Carbon storage under LP+FP scenario (2050)

#### 9.7.8.3 8.3 Pollination

Models pollinator abundance and crop pollination dependency using InVEST's Pollination model.

**Key Output:**

```
intermediate/run_invest_pollination/
   esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050/
       total_pollinator_abundance_summer.tif    # Pollinator index
```

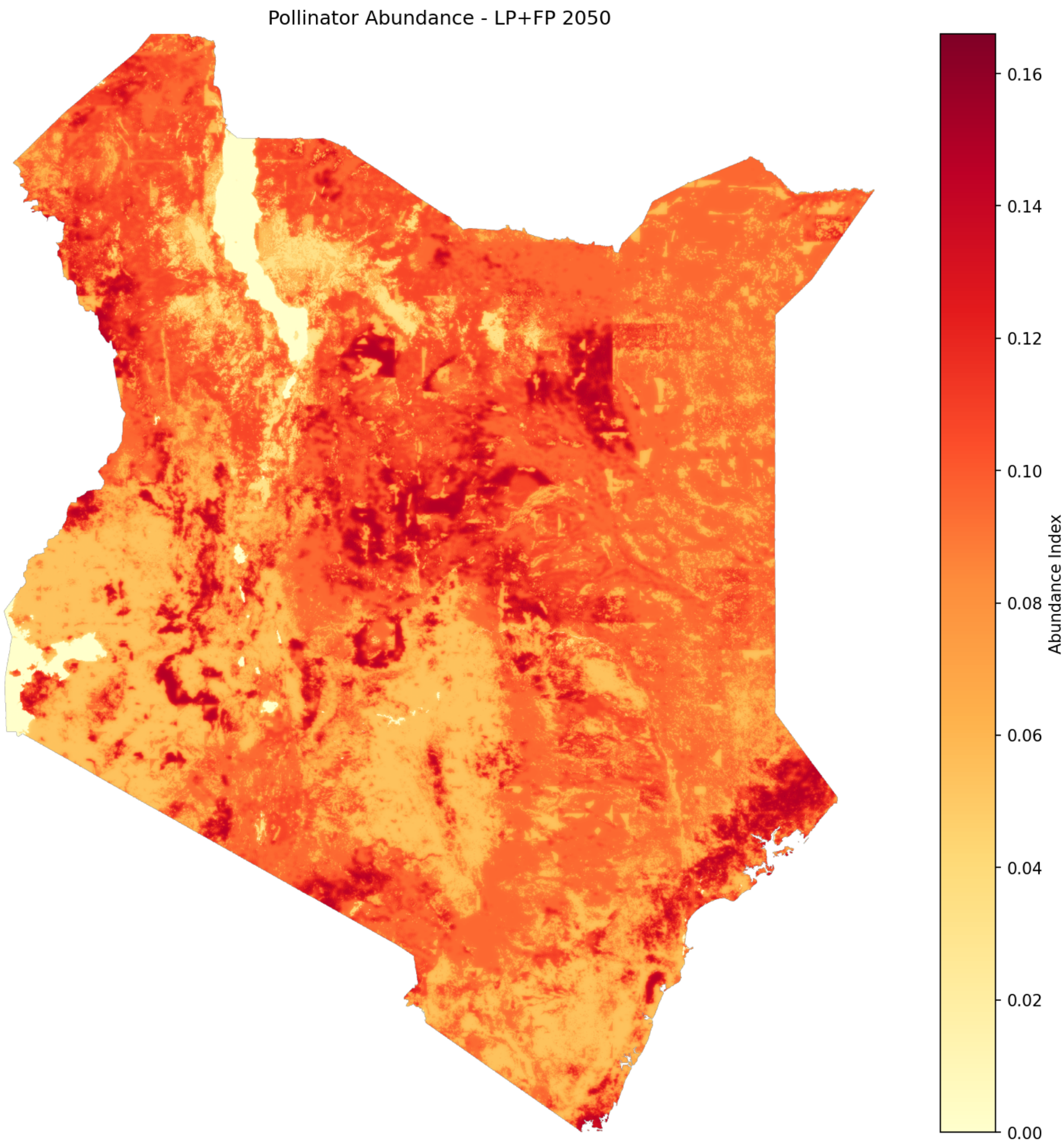



Figure 9.11: Pollinator abundance under LP+FP scenario (2050)

#### 9.7.8.4 8.4 Crop Value and Pollination Shocks

Calculates agricultural value and applies DEMETRA economic shocks based on pollination changes.

**What it does:**

1. Baseline crop value: Combines production data from all crops weighted by pollination dependency
2. Maximum potential loss: Calculates loss if pollination disappeared completely
3. Pollination sufficiency threshold: Uses 0.3 as the critical threshold
4. Adjusted crop value: For each scenario:
    - If below threshold → apply proportional loss
    - If at or above threshold → maintain baseline value
5. Regional shocks: Calculates `shock_value = scenario_value / baseline_value` by region

**Key Outputs:**

```
intermediate/calculate_crop_value_and_shock/
  crop_value_baseline.tif                      # Baseline crop value ($/ha)
  crop_value_max_lost.tif                      # Maximum possible loss ($/ha)
  crop_value_adjusted_baseline_2020.tif        # Adjusted baseline
  crop_value_adjusted_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050.tif  # Scenario
  pollination_shocks_seals7_ssp2_rcp45_luh2-message.csv          # Regional shock values
  pollination_shocks_alt_seals7_ssp2_rcp45_luh2-message.csv      # Alternative regions
```

**Significance:**

- Translates ecological changes (pollination) into economic impacts
- Provides spatially explicit economic loss estimates
- Generates shock values for DEMETRA CGE model integration

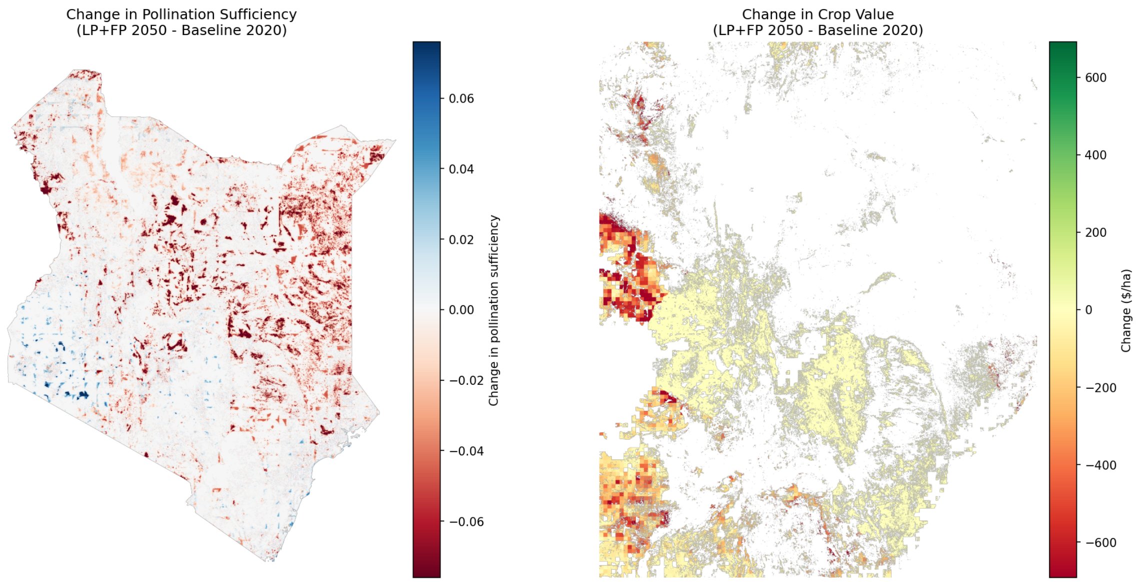



Figure 9.12: Pollination change and Crop value change

Table 9.1: Region Pollination Shocks (BAU, 2050)

| Year | Region | Shock_Value |
|---|---|---:|
| 2050 | AN | 0.86908 |
| 2050 | AS | 0.919913 |
| 2050 | CO | 0.921871 |
| 2050 | HR | 0.924216 |
| 2050 | MN | 0.924557 |
| 2050 | MO | 0.929781 |
| 2050 | MS | 0.977373 |
| 2050 | NA | 0.99532 |

#### 9.7.8.5 8.5 Biodiversity Index

Calculates habitat quality and biodiversity index based on LULC patterns and biodiversity data (species ranges, endemic species ranges, key biodiversity areas, ecoregions). Not currently an official part of the InVEST ecosystem.

**Key Output:**

```
intermediate/calculate_biodiversity_index/
  esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050/
```

`esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050_index.tif`

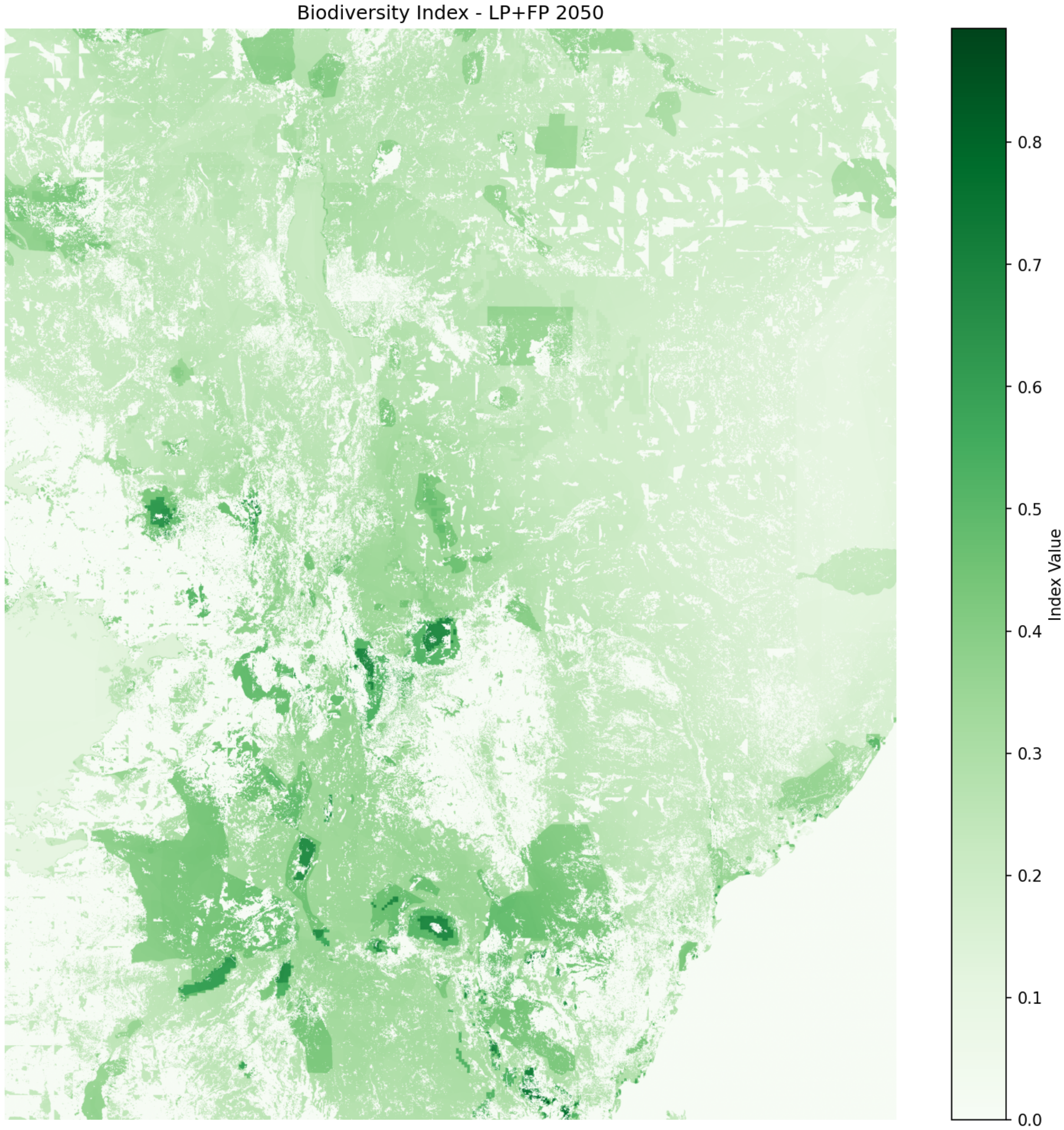



Figure 9.13: Biodiversity index under LP+FP scenario (2050)

### 9.7.9 9. Summary and Visualization

**Significance:** Creates multi-panel comparison figures showing results across all scenarios and ecosystem services.

**What it does:**

1. Compiles results from all InVEST modules
2. Generates comparison figures across scenarios (bau, LP, FP, LP+FP, enablers, es_index)
3. Creates summary visualizations for reports and presentations

**Key Outputs:**

```
intermediate/summarize_and_visualize_multi_vector/
  biodiversity_combined/
     esa_seals7_ssp2_rcp45_luh2-message_bau_biodiversity_combined.png
  carbon_combined/
     esa_seals7_ssp2_rcp45_luh2-message_es_index_carbon_combined.png
  lulc_combined/
     esa_seals7_ssp2_rcp45_luh2-message_enablers_lulc_combined.png
  pollination_combined/
     esa_seals7_ssp2_rcp45_luh2-message_es_index_pollination_combined.png
  summary_visualizations/
      summary_grouped_vertical_2050.png        # Master summary figure
```

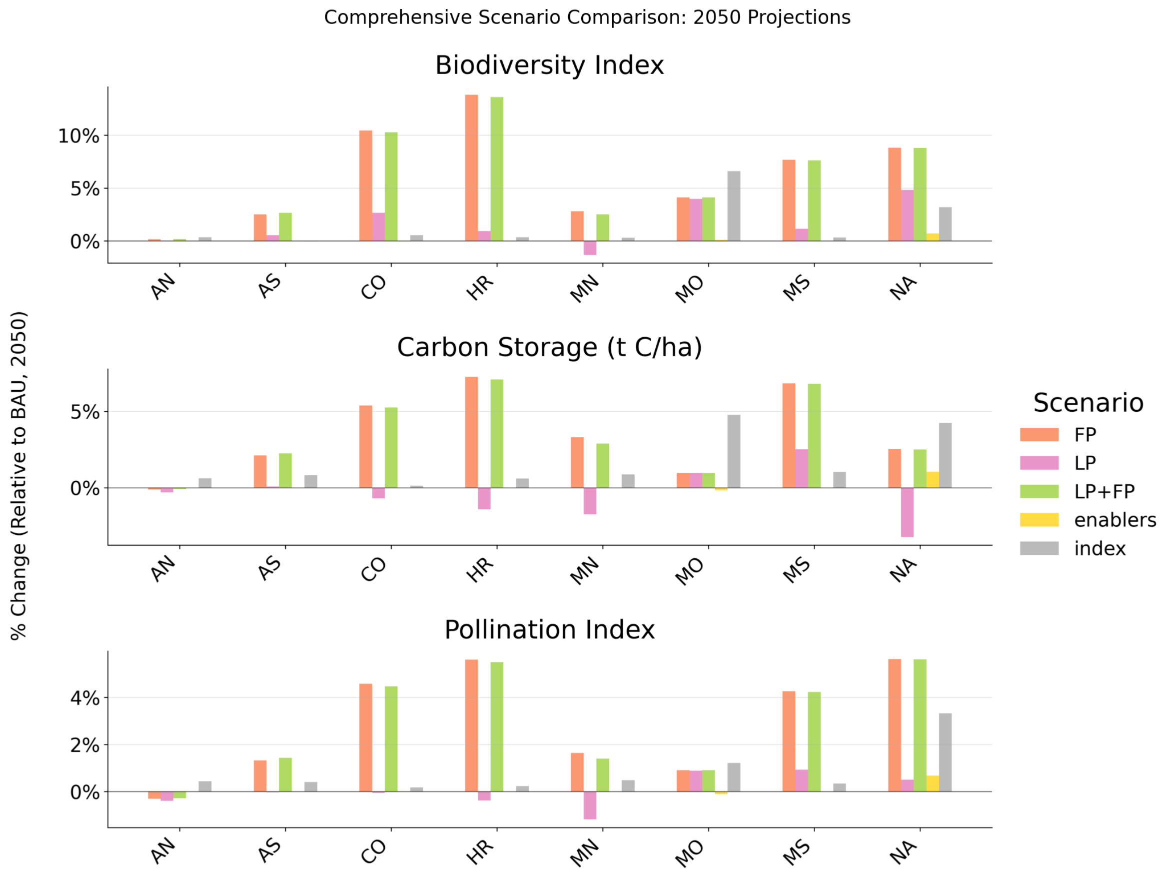



Figure 9.14: Master summary visualization comparing all scenarios (2050)

---

## 9.8 Final Outputs

Output files will be generated in:

```
<home>/Files/seals/projects/run_ken_demetra/
```

**Primary outputs include:**

**SEALS outputs:**

- LULC geotiff files: `intermediate/stitched_lulc_simplified_scenarios/`

    - Land-use/land-cover predictions for each scenario year

- LULC png files: `intermediate/visualization/lulc_pngs/`
    - Spatial representation of land-use transitions

**Project outputs:**

- Figures: `intermediate/summarize_and_visualize_multi_vector/`
    - `biodiversity_combined`
    - `carbon_combined`
    - `carbon_total_combined`
    - `lulc_combined`
    - `pollination_combined`
    - `summary_visualizations`

---

## 9.9 Additional Resources

- SEALS Repository: https://github.com/jandrewjohnson/seals
- InVEST Documentation: https://naturalcapitalalliance.stanford.edu/software/invest
- InVEST Python Documentation: https://invest.readthedocs.io/en/latest/index.html

# 10 Release Notes

Release history for SEALS. Versions continue this repository's own tags (v1.2.0 preceded v2.0.0).

For the shared platform beneath SEALS — hazelbean, ProjectFlow and the conventions — see the Earth-Economy Devstack release notes.

## 10.1 SEALS v2.0.0 (2026-07-30)

**Run-file spec: three complexity axes, and run_project(p).**

- A run file's complexity is now described by three *independent* axes rather than one ladder. **Configuration** (how a run varies) keeps the numbered levels 1–4: one task with inline constants; a task tree; + a scenarios CSV; + a parameters CSV. **Code layout** (where the code lives) runs single file → split (<project>_tasks.py, _functions.py, _utils.py, _initialize_project.py) → library package. **Ownership** (who owns the code you run) runs self-contained → devstack developer → downstream user. Moving along one axis never requires moving along another, so *"level" unqualified always means the configuration axis.*

- run_project now takes only p. The caller builds and configures the ProjectFlow in the __main__ guard; run_project(p) only runs it. The rule that replaces a signature full of keyword defaults: **run_project(p) sets what no variant ever changes; the caller sets what a variant might.** A constant that starts varying moves one line up instead of becoming a new keyword argument, so the entry point stops growing as a project accumulates definition CSVs. Cost: bare run_project() no longer works, and a variant wrapper is four lines rather than three.

- <project>_utils.py is defined as a **promotion queue** into hazelbean — science-unaware helpers used by more than one project belong upstream — while <project>_functions.py holds the model's domain knowledge and is terminal.

- examples/ reorganized: run_templates/ (copy-me) and run_templates_annotated/ (the same code with the reasoning written out), plus input_research_scripts/ for real run files and the raw scripts they replace. Two new templates hold configuration fixed at level 4 and move one other axis each: run_template_split_layout.py and run_template_downstream_user.py.

- The six numbered sections of *project_complexity.qmd* are renamed to "Stage N, the old way" / "Stage N, redone with ProjectFlow" so they no longer collide with the configuration levels. They are a historical narrative, not the spec.
- All 46 `*_old_spec.py` archives are deleted across the stack. Git history is the archive for pre-conversion forms. The empty `docs/run_file.qmd` stub is removed; `examples/index.qmd` is the landing page for "what should my run file look like?".